Supplementary Material for When Does Auxiliary Data Help? Global Geometry and Optimal Mixing under Task Mismatch

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The main text follows three stages: two-dimensional multivalley geometry, global population certification, and finite-sample structure and selection. For the core arguments, read Sec. M for population geometry and certificates, Sec. K for the full-ridge risk expansion, Sec. L for uniform derivatives and interior optima, and Sec. N for flat and multiple optima. Section O collects further consequences for matched-spectrum gain, a first-order exchange equation, and a quadratic-surrogate dosage rule. The earlier sections retain the diagonalized reference model and its separate risk objects; C1 denotes its marginal-benefit decomposition and C2 its marginal criterion.

Table 1. Claim–object map. Exact identities for the diagonalized leading-order object are separate from asymptotic expansions of the actual full-ridge risk.

Result	Estimator and risk object	Established scope
Population shape	Full ridge; population risk	Zero-mismatch log-convexity, minimum-dimension multimodality, and fixed-parameter polynomial global certification
T1-c, full derivative	Standard full ridge; actual risk	Fixed-p second-order risk and uniform first/second derivatives; zero-mixing expansion is a special case
Interior optimum	Standard full ridge; actual risk	First-order optimal-weight correction and optimized risk/gain expansions at a unique nondegenerate interior population minimizer; no boundary-switching claim
Boundary optima	Standard full ridge; actual risk	Nondegenerate boundary persistence and flat-boundary weight/gain rates; local–global counterexample included
Multiple optima	Full ridge; population and first order	Valley bifurcations, two global minimizers, and R_1 branch selection at first order
Flat R_0	Full ridge; matched spectra and $\delta = 0$	Exact $R_1 = A(1-w)^2 + Bw^2$; if $A + B = 0$, actual risk and gain vanish identically and every weight is optimal
Matched gain, exchange	Full ridge; matched spectra	Optimized-gain expansion and exchange equation based on first-order risks
Surrogate dosage	Population-plus-noise quadratic	Exact piecewise optimum of this surrogate; no full-ridge dosage guarantee
T1-a	Diagonalized; leading order	Conditional per-coordinate approximation, not an exact finite-sample identity
C1, C2	Diagonalized; $\bar{\mathcal{R}}_T^{\mathrm{lead}}$	Exact marginal decomposition and critical-size algebra for this object
Exchange rate, dosage	Diagonalized; $\bar{\mathcal{R}}_T^{\mathrm{lead}}$	Full-interval optimization under the stated baseline and spectral conditions

A. Setup and notation

We restate the model of the main text (Sec. 3) for self-containment. Coordinates $j = 1, \dots, p$ index a shared spectral basis; domain $e \in \{r, h\}$ has covariates $x_e = \Sigma_e^{1/2} u_e$, u_e standard Gaussian, $\Sigma_e = \mathrm{diag}(\lambda_j^e)$ with $\lambda_j^e \asymp j^{-\alpha_e}$, $\lambda_1^e = 1$; we write $a_j = \lambda_j^r$, $b_j = \lambda_j^h$. Labels: $y_r = \langle \beta^*, x_r \rangle + \varepsilon_r$, $y_h = \langle \beta^* + \delta, x_h \rangle + \varepsilon_h$; $\sum_j j^{2s} \beta_j^{*2} \le 1$; noise variances σ_e^2. Given n_r robot and n_h human samples, mixing ratio $\lambda \in [0, 1]$, per-sample weights $w_r = (1-\lambda)/n_r$, $w_h = \lambda/n_h$, the diagonalized ridge estimator is

$$\hat{\beta}_j = \frac{w_r T_{rj} + w_h T_{hj}}{w_r S_{rj} + w_h S_{hj} + \mu}, \qquad S_{ejk} = \textstyle\sum_{i=1}^{n_e} x_{e,ij} x_{e,ik}, \quad T_{ej} = \sum_i x_{e,ij}\, y_{e,i}. \tag{1}$$

The robot-domain excess risk is $\mathcal{R}_T(\lambda) = \mathbb{E}\|\hat{\beta}(\lambda) - \beta^*\|_{\Sigma_r}^2 = \sum_j a_j\, \mathbb{E}(\hat{\beta}_j - \beta_j^*)^2$. For this diagonalized estimator, the coordinatewise effective noise is $\tilde{\sigma}_{e,j}^2 = \sigma_e^2 + F_{e,j}$ with per-coordinate feedthrough energy $F_{e,j} = \sum_{k \ne j} \lambda_k^e \beta_{e,k}^2$ ($\beta_h = \beta^* + \delta$). It describes the signal terms left in a coordinate numerator when the estimator discards off-diagonal Gram entries; it is not an additional independent noise source for full ridge. Where the coordinate dependence is immaterial we abbreviate $\tilde{\sigma}_{e,j}^2$ (resp. $F_{e,j}$) as $\tilde{\sigma}_e^2$ (resp. F_e), and numerical evaluations of this diagonalized object use the exact per-coordinate form.

Single-coordinate notation (used throughout, index j suppressed). $U_r \sim \chi_{n_r}^2$, $U_h \sim \chi_{n_h}^2$ independent; $S_e = \lambda^e U_e$, so $\mathbb{E}S_e = n_e \lambda^e$, $\mathrm{Var}(S_e) = 2n_e(\lambda^e)^2$, and the third central moment is $8n_e(\lambda^e)^3$. $\hat{D} = w_r S_r + w_h S_h + \mu$; $D = \mathbb{E}\hat{D} = (1-\lambda)a +$

$\lambda b+\mu$. $\hat{N} = w_r T_r + w_h T_h$; its systematic part $N_0 = \lambda b\,\delta - \mu\beta^*$. The fluctuation scale is $\eta = \sqrt{8\log(4p/\delta_c)}\,\big(\frac{(1-\lambda)a}{\sqrt{n_r}} + \frac{\lambda b}{\sqrt{n_h}}\big)$ for confidence parameter $\delta_c \in (0,1)$. We write $\tilde{n} = \hat{N} - \mathbb{E}\hat{N}$; direct expansion gives

$$\mathrm{Var}(\tilde{n}) = \underbrace{\frac{2\lambda^2(b\delta)^2}{n_h}}_{\text{mismatch fluctuation}} + \underbrace{\frac{(1-\lambda)^2\tilde{\sigma}_r^2 a}{n_r} + \frac{\lambda^2\tilde{\sigma}_h^2 b}{n_h}}_{\text{noise + feedthrough}}. \tag{2}$$

B. Lemmas L1 and L2: per-coordinate representation and conditional MSE

Lemma 1 (L1, representation)**.** *For each coordinate j, with the definitions above,*

$$T_{ej} = S_{ejj}\,\beta_{e,j} + U_{ej} + \textstyle\sum_i x_{e,ij}\varepsilon_{e,i}, \qquad U_{ej} = \textstyle\sum_{k\neq j} S_{ejk}\,\beta_{e,k}, \tag{3}$$

where U_{ej} is mean-zero with $\mathrm{Var}(U_{ej}) = n_e\lambda_j^e F_{e,j}$, $F_{e,j} = \sum_{k\neq j}\lambda_k^e\beta_{e,k}^2$.

Proof. Expand $y_{e,i} = \sum_k x_{e,ik}\beta_{e,k} + \varepsilon_{e,i}$ in $T_{ej} = \sum_i x_{e,ij}y_{e,i}$ and separate the $k = j$ term. For $k \neq j$, $\mathbb{E}S_{ejk} = \sum_i \mathbb{E}[x_{e,ij}]\mathbb{E}[x_{e,ik}] = 0$ by independence of distinct coordinates of the Gaussian covariate, hence $\mathbb{E}U_{ej} = 0$. For the variance, $\mathrm{Var}(S_{ejk}) = \sum_i \mathbb{E}[x_{e,ij}^2]\mathbb{E}[x_{e,ik}^2] = n_e\lambda_j^e\lambda_k^e$ (independence of coordinates $j \neq k$), and distinct cross terms are uncorrelated: for $k \neq l$, $\mathrm{Cov}(S_{ejk}, S_{ejl}) = \sum_i \mathbb{E}[x_{e,ij}^2 x_{e,ik}x_{e,il}] = 0$ since $\mathbb{E}[x_{e,ik}x_{e,il}] = 0$ and $x_{e,ij}^2$ is independent of $(x_{e,ik}, x_{e,il})$. Summing, $\mathrm{Var}(U_{ej}) = \sum_{k\neq j}\beta_{e,k}^2\, n_e\lambda_j^e\lambda_k^e = n_e\lambda_j^e F_{e,j}$. We retain the exact per-coordinate form $F_{e,j}$ throughout; no approximation of $F_{e,j}$ by the total signal energy $\sum_k \lambda_k^e\beta_{e,k}^2$ is made, since the omitted single term need not be negligible — for a low-frequency-concentrated β^* (the regime of the ellipsoid constraint in the main text), the first coordinates can carry an $O(1)$ fraction of the total energy. □

Remark 1. The feedthrough U_{ej} contributes variance $n_e\lambda_j^e F_{e,j}$ to the coordinate-j numerator, which motivates the effective noise $\tilde{\sigma}_{e,j}^2$ in Eq. (2). This is a variance calculation only: U_{ej} and S_{ejj} share $x_{e,ij}$ and are not independent. In fact, $\mathrm{Var}(U_{ej} \mid X_{e,:j}) = F_{e,j}S_{ejj}$. Higher-moment arguments must not treat them as independent.

Lemma 2 (L2, conditional MSE)**.** *Define $N_{0j}^X = w_h S_{hjj}\delta_j - \mu\beta_j^* + w_r U_{rj} + w_h U_{hj}$, which is deterministic conditional on X. Then*

$$\mathbb{E}\big[(\hat{\beta}_j - \beta_j^*)^2 \mid X\big] = \frac{(N_{0j}^X)^2}{\hat{D}_j^2} + \frac{w_r^2\sigma_r^2 S_{rjj}}{\hat{D}_j^2} + \frac{w_h^2\sigma_h^2 S_{hjj}}{\hat{D}_j^2}. \tag{4}$$

Proof. Substitute Lemma 1 into Eq. (1): $\hat{\beta}_j\hat{D}_j = w_r S_{rjj}\beta_j^* + w_h S_{hjj}(\beta_j^* + \delta_j) + (\text{feedthrough} + \text{noise})$. Subtract $\beta_j^*\hat{D}_j = w_r S_{rjj}\beta_j^* + w_h S_{hjj}\beta_j^* + \mu\beta_j^*$. The systematic numerator is N_{0j}^X. The remaining zero-mean label-noise projection is $w_r X_{r,:j}^\top\varepsilon_r + w_h X_{h,:j}^\top\varepsilon_h$. Its conditional variance is $w_r^2\sigma_r^2 S_{rjj} + w_h^2\sigma_h^2 S_{hjj}$, and cross terms vanish by independence and zero mean. □

C. Lemma L3: leading per-coordinate approximation

Lemma 3 (L3, leading order)**.** *Fix a coordinate and suppose $D \geq 2\eta$ (information-rich regime). Define*

$$\mathrm{MSE}^{\mathrm{lead}} = \frac{N_0^2 + \mathrm{Var}(\tilde{n})}{D^2}, \qquad \overline{N}^2 = N_0^2 + \mathrm{Var}(\tilde{n}), \qquad M_4 = (\mathbb{E}\hat{N}^4)^{1/2}. \tag{5}$$

Then

$$\left|\mathbb{E}\left[\frac{\hat{N}^2}{\hat{D}^2}\right] - \mathrm{MSE}^{\mathrm{lead}}\right| \leq C\,\frac{\eta}{D}\,\frac{M_4}{D^2} + O\!\left(\frac{\eta^2}{D^2}\,\frac{\overline{N}^2}{D^2}\right) + O\Big(\sqrt{\delta_c}\,\mu^{-2}M_4\Big), \tag{6}$$

where C is an absolute constant and M_4 is finite under the Gaussian design and the noise moment assumptions.

Proof. Step 1 (concentration). Write $\Delta = \hat{D} - D$. By the Laurent–Massart χ^2 tail bound and a union bound, there is an event G with $\Pr(G) \geq 1 - \delta_c$ on which $|\Delta| \leq \eta \leq D/2$ simultaneously for all retained coordinates. Also, $\mathbb{E}\Delta^2 = O(\eta^2)$.

Step 2 (Taylor expansion on G). For $f(x) = x^{-2}$,

$$f(D + \Delta) = D^{-2} - 2\Delta D^{-3} + R, \qquad |R| \leq 48\eta^2/D^4. \tag{7}$$

Splitting the expectation into good and bad events gives

$$\begin{aligned}\mathbb{E}[\hat{N}^2 f(\hat{D})] &= \mathbb{E}[\hat{N}^2 f(\hat{D})\mathbf{1}_G] + \mathbb{E}[\hat{N}^2 f(\hat{D})\mathbf{1}_{G^c}] \\ &= D^{-2}\mathbb{E}\hat{N}^2 - 2D^{-3}\mathbb{E}[\hat{N}^2\Delta\mathbf{1}_G] \\ &\quad + \mathbb{E}[\hat{N}^2 R\mathbf{1}_G] + \mathbb{E}[\hat{N}^2 f(\hat{D})\mathbf{1}_{G^c}].\end{aligned} \tag{8}$$

The difference from $\overline{N}^2/D^2$ therefore contains four terms: the subtracted $D^{-2}\mathbb{E}[\hat{N}^2\mathbf{1}_{G^c}]$, the cross term $-2D^{-3}\mathbb{E}[\hat{N}^2\Delta\mathbf{1}_G]$, the Taylor remainder $\mathbb{E}[\hat{N}^2 R\mathbf{1}_G]$, and the actual bad-event risk $\mathbb{E}[\hat{N}^2 f(\hat{D})\mathbf{1}_{G^c}]$.

Cauchy–Schwarz gives

$$\begin{aligned}|\mathbb{E}[\hat{N}^2\Delta\mathbf{1}_G]| &\leq (\mathbb{E}\hat{N}^4)^{1/2}(\mathbb{E}\Delta^2)^{1/2} \\ &= O(\eta M_4).\end{aligned} \tag{9}$$

The Taylor term is at most $48\eta^2\overline{N}^2/D^4$. No independence between feedthrough and $\hat{D}$ is used.

Step 3 (both complementary-event terms). Since $\hat{D} \geq \mu > 0$, another Cauchy–Schwarz bound yields

$$\begin{aligned}D^{-2}\mathbb{E}[\hat{N}^2\mathbf{1}_{G^c}] &\leq D^{-2}\sqrt{\delta_c}\, M_4, \\ \mathbb{E}[\hat{N}^2 f(\hat{D})\mathbf{1}_{G^c}] &\leq \mu^{-2}\sqrt{\delta_c}\, M_4.\end{aligned} \tag{10}$$

Combining the four terms proves Eq. (6); the first two are controlled by $C(\eta/D)M_4/D^2$ because $\eta/D \leq 1/2$. □

A corrected numerical surrogate, not a second-order theorem. For computation, we also evaluate

$$\begin{aligned}\mathrm{MSE}^{\mathrm{det}} = \mathrm{MSE}^{\mathrm{lead}} &- \frac{4N_0\mathrm{Cov}(\tilde{n}, \hat{D})}{D^3} \\ &+ \frac{3N_0^2\mathrm{Var}(\hat{D})}{D^4},\end{aligned} \tag{11}$$

with $\mathrm{Cov}(\tilde{n}, \hat{D}) = 2\lambda^2\delta b^2/n_h$ and $\mathrm{Var}(\hat{D}) = 2[(1-\lambda)^2 a^2/n_r + \lambda^2 b^2/n_h]$. This is a delta-method-inspired numerical surrogate. It is not covered by Lemma 3: feedthrough and the denominator are dependent, and the mixed moments and higher-order terms required for a uniform corrected error bound are not asserted to be controlled. In particular, we do not claim that the mixed moment $\mathbb{E}[\tilde{n}^2\Delta]$ or all higher-order numerator terms vanish.

D. Aggregation and truncation scope

No global truncation lemma is asserted under Assumptions A1–A3. The ellipsoid constraint $\sum_j j^{2s}\beta_j^{*2} \leq 1$ does not imply a coordinatewise law $\beta_j^{*2} \asymp j^{-(2s+1)}$ or a two-sided power-law tail. It permits sparse and concentrated task vectors. Moreover, a truncated coordinate can have $\beta_j^* = 0$ but $\delta_j \neq 0$, in which case human data can produce nonzero bias and variance; its risk cannot be $\Theta(\beta_j^{*2})$. Even with zero mismatch, noise can prevent such an equivalence. Consequently, we do not claim a global error bound obtained by replacing the tail with $\sum_{j>m} a_j\beta_j^{*2}$, nor do we assert a tail rate or a boundary-band cardinality without additional assumptions. A global approximation to the true total risk requires separate control of task mismatch, noise, and the information-poor coordinates.

E. Two deterministic risk objects and the scope of exact identities

Define the leading-order and corrected per-coordinate risks by

$$\begin{aligned}\mathrm{MSE}^{\mathrm{lead}} &:= \frac{N_0^2 + \mathrm{Var}(\tilde{n})}{D^2},\\ \mathrm{MSE}^{\mathrm{det}} &= \mathrm{MSE}^{\mathrm{lead}} - \frac{4N_0\mathrm{Cov}(\tilde{n},\hat{D})}{D^3}\\ &\quad + \frac{3N_0^2\mathrm{Var}(\hat{D})}{D^4}.\end{aligned} \tag{12}$$

The leading numerator includes the mismatch-fluctuation term in Eq. (2). Aggregating the per-coordinate expressions over the explicitly specified retained set in each protocol defines the formal objects $\bar{\mathcal{R}}_T^{\mathrm{lead}}$ and $\bar{\mathcal{R}}_T^{\mathrm{det}}$, respectively. These are deterministic risk objects used for analysis and computation, not a proved truncation approximation to the true total risk. All structural identities below (the marginal decomposition, the H criterion, unimodality, the matched-spectrum quadratic form, the exchange rate, and the dosage law) concern $\bar{\mathcal{R}}_T^{\mathrm{lead}}$. The L3 leading-order approximation concerns $\bar{\mathcal{R}}_T^{\mathrm{lead}}$; corrected numerical risk curves use the surrogate $\bar{\mathcal{R}}_T^{\mathrm{det}}$.

Proposition 4 (explicit derivative correction at zero mixing)**.** *Write $u = b\delta$, $w = \mu\beta^*$, and $D_0 = a + \mu$. Then, coordinate by coordinate,*

$$\begin{aligned}\partial_\lambda \mathrm{MSE}^{\mathrm{det}}\big|_{0^+} &= \partial_\lambda \mathrm{MSE}^{\mathrm{lead}}\big|_{0^+} - \frac{12a^2w(u+w)}{n_r D_0^4}\\ &\quad - \frac{24a^2w^2(b-a)}{n_r D_0^5}.\end{aligned} \tag{13}$$

The covariance correction contributes zero at $\lambda = 0$. The aggregate correction is obtained by multiplying the displayed terms by a_j and summing over retained coordinates.

Proof. Put $C(\lambda) = \mathrm{Cov}(\tilde{n},\hat{D}) = 2\lambda^2\delta b^2/n_h$ and $W(\lambda) = \mathrm{Var}(\hat{D}) = 2[(1-\lambda)^2a^2/n_r + \lambda^2b^2/n_h]$. Since $C(0) = C'(0) = 0$, differentiating $-4N_0C/D^3$ at zero gives no contribution. Moreover $N_0(0) = -w$, $N_0'(0) = u$, $W(0) = 2a^2/n_r$, $W'(0) = -4a^2/n_r$, and $D'(0) = b - a$. Differentiating $3N_0^2W/D^4$ and substituting these values gives Eq. (13). □

Thus the correction is $O(n_r^{-1})$ per coordinate and vanishes when $n_r \to \infty$ with n_h/n_r fixed: the asymptotic functional H is unchanged. Finite-n_r statements such as $n_r^{\mathrm{crit}} = K/H$ are exact for $\bar{\mathcal{R}}_T^{\mathrm{lead}}$; for $\bar{\mathcal{R}}_T^{\mathrm{det}}$ the additional terms in Eq. (13) must be included instead of treating K as correction-free. Likewise, at matched spectra N_0 is affine and C, W are quadratic, so $\mathrm{MSE}^{\mathrm{det}}$ generally contains cubic and quartic terms in λ. The exact quadratic and hyperbolic dosage identities below therefore apply to $\bar{\mathcal{R}}_T^{\mathrm{lead}}$, not to $\bar{\mathcal{R}}_T^{\mathrm{det}}$ or to the unapproximated finite-sample risk.

F. Proof of the marginal-benefit decomposition (C1)

Differentiate the leading-order expression in Eq. (12) at $\lambda = 0^+$ per coordinate. With $D_0 = a + \mu$, $V = \tilde{\sigma}_{r,j}^2 a/n_r$, direct expansion gives

$$\partial_\lambda \mathrm{MSE}^{\mathrm{lead}}\big|_{0^+} = \frac{-2\mu\beta^* b\,\delta - 2V}{D_0^2} - \frac{2\big(V + \mu^2\beta^{*2}\big)(b-a)}{D_0^3}, \tag{14}$$

which, after flipping the sign and dropping the positive factor $2/D_0^3$, is the three-term decomposition of the main text:

$$\mathrm{benefit}_j \;\propto\; \underbrace{\mu\beta_j^* b_j \delta_j D_{0j}}_{\text{A: alignment}} + \underbrace{V_j D_{0j}}_{\text{B: variance}} + \underbrace{\big(V_j + \mu^2\beta_j^{*2}\big)(b_j - a_j)}_{\text{C: coverage}}. \tag{15}$$

Analytic derivative vs. finite differences on the leading-order formula: relative error $< 1\%$ (configuration of the v2 benchmark). The derivative of $\mathrm{MSE}^{\mathrm{det}}$ has the additional explicit terms of Proposition 4. At $\mu = 0$, $D_0 = a$, terms A and the shrinkage part of C vanish, and $\partial_\lambda \bar{\mathcal{R}}_T^{\mathrm{lead}}(0^+)|_{\mu=0} = -(2/n_r)\sum_j \tilde{\sigma}_{r,j}^2 b_j/a_j < 0$ always. For exact power-law spectra and coordinate-invariant effective noise, this reduces to $-(2\tilde{\sigma}_r^2/n_r)\sum_j j^{\alpha_r - \alpha_h}$.

G. The central H-criterion: derivation and scope

Take $n_r \to \infty$ with n_h/n_r fixed in Eq. (14): the variance term B vanishes, and the zero-mixing risk derivative converges to H (equivalently, the marginal benefit converges to $-H$), for a general mismatch vector:

$$H = \underbrace{-2\mu \sum_j \frac{a_j \beta_j^* b_j \delta_j}{D_{0j}^2}}_{A_\infty} \underbrace{- 2\mu^2 \sum_j \frac{a_j \beta_j^{*2}(b_j - a_j)}{D_{0j}^3}}_{C_\infty}. \tag{16}$$

At finite n_r the variance terms of $\bar{\mathcal{R}}_T^{\text{lead}}$ contribute $-K/n_r$ to the derivative, with $K = 2\sum_j \tilde{\sigma}_{r,j}^2 (S_{1j} + S_{2j})$, $S_{1j} = a_j^2/D_{0j}^2$, $S_{2j} = a_j^2(b_j - a_j)/D_{0j}^3$; hence marginal mixing is beneficial iff $K/n_r > H$, giving $n_r^{\text{crit}} = K/H$ when $H > 0$. Here $S_{1j} + S_{2j} = a_j^2(b_j + \mu)/D_{0j}^3 > 0$, so $K > 0$ whenever $\tilde{\sigma}_{r,j}^2 > 0$. Proposition 4 vanishes in this large-n_r limit, so H is unchanged for $\bar{\mathcal{R}}_T^{\text{det}}$; the displayed finite-$n_r$ formula is exact for $\bar{\mathcal{R}}_T^{\text{lead}}$, with the stated correction required for $\bar{\mathcal{R}}_T^{\text{det}}$. Under the experimental convention $\beta_j^* > 0$ and $\delta_j = -z_j$ with $z_j \geq 0$, the signed first term reduces to $2\mu \sum_j a_j \beta_j^* b_j z_j / D_{0j}^2$; hence the anti-aligned numerical configurations also satisfy the earlier absolute-value specialization. Numerically (v11, $p = 1000$, $\mu = 0.05$, spectral family A): A_∞ is linear in $\|\delta\|$ with slope 6.975×10^{-3}, $C_\infty = 4.700 \times 10^{-3}$ independent of δ, predicting $\|\delta\|^* = 0.674$, which brackets the sign change of the analytic/numerical risk derivative; and $n_r^{\text{crit}} \cdot H = 25.866$ is constant across $\|\delta\| \in \{0.75, 1.0, 2.0\}$ to display precision when both sides are computed from the same leading-order formula. These v11 checks are algebraic/numerical self-consistency, not independent Monte-Carlo phase validation.

The classical limit and its order of limits. At fixed finite dimension, $H = O(\mu)$ as $\mu \to 0$, and if $A_\infty > 0$ then $n_r^{\text{crit}} = K/H \to \infty$. Thus, at any fixed n_r, sufficiently small μ gives $K/n_r > H$ and marginal mixing is beneficial, consistently with the $\mu = 0$ identity above. The discard-the-source transition is recovered only when n_r crosses the threshold as μ decreases (asymptotically $n_r \gtrsim 1/\mu$, with constants determined by K/A_∞); it is not obtained by taking $\mu \to 0$ at fixed n_r. At fixed $\mu > 0$ the large-n_r marginal mixing benefit tends to $-A_\infty + C_\infty = -H$: when $H < 0$, regularization preserves a positive coverage benefit even with unlimited robot data.

H. Consequences: exchange rate, unimodality, and dosage

Throughout this section $\bar{\mathcal{R}}_T$ denotes $\bar{\mathcal{R}}_T^{\text{lead}}$; the denominator-fluctuation corrections of $\bar{\mathcal{R}}_T^{\text{det}}$ are not part of these algebraic identities. The exchange rate is defined implicitly by $\bar{\mathcal{R}}_T(\lambda^*;\, n_r, n_h) = \bar{\mathcal{R}}_T(0;\, n_r + \kappa_{\text{ex}} n_h)$, where λ^* minimizes the mixture risk over $[0, 1]$. This section proves that κ_{ex} is well-defined without any unimodality assumption (Proposition 5), that every per-coordinate risk is unimodal with an explicit minimizer (Lemma 7 and Proposition 8), and that global unimodality holds exactly in the matched-spectrum regime (Theorem 9) and under an explicit convexity condition in general (Proposition 10).

H.1. Well-definedness of the exchange rate

Proposition 5 (existence and uniqueness of κ_{ex})**.** *Fix (n_r, n_h) and write $m_\star = \min_{\lambda \in [0,1]} \bar{\mathcal{R}}_T(\lambda; n_r, n_h)$ (attained: $\bar{\mathcal{R}}_T$ is continuous in λ on a compact set). As a function of the robot sample size n at $\lambda = 0$,*

$$\bar{\mathcal{R}}_T(0; n) \;=\; \sum_{j > m} a_j \beta_j^{*2} \;+\; \sum_{j \leq m} a_j \, \frac{\tilde{\sigma}_{r,j}^2 a_j / n + \mu^2 \beta_j^{*2}}{(a_j + \mu)^2} \tag{17}$$

is continuous and strictly decreasing in n whenever $\tilde{\sigma}_{r,j}^2 > 0$ for every retained coordinate (in particular whenever $\beta^* \neq 0$, since $\tilde{\sigma}_{r,j}^2 = \sigma_r^2 + F_{r,j} > 0$), with limit

$$\bar{\mathcal{R}}_T(0; \infty) \;=\; \sum_j a_j \, \frac{\mu^2 \beta_j^{*2}}{(a_j + \mu)^2}, \tag{18}$$

the pure shrinkage risk (the truncation recedes, $m \to p$, as $n \to \infty$). Since $\lambda = 0$ is feasible, $m_\star \leq \bar{\mathcal{R}}_T(0; n_r)$, and therefore: (i) if $m_\star > \bar{\mathcal{R}}_T(0; \infty)$, there exists a unique finite $\kappa_{\text{ex}} \geq 0$ solving the defining equation; (ii) if $m_\star \leq \bar{\mathcal{R}}_T(0; \infty)$, no finite solution exists and we write $\kappa_{\text{ex}} = +\infty$. Values greater than one are ordinary finite exchange rates; no truncation at one is part of the definition. If a capped display is desired, it must be reported separately as $\kappa_{\text{ex}}^{\text{cap}} = \min(\kappa_{\text{ex}}, 1)$. Only the minimum value *$m_\star$ enters the construction; unimodality of $\bar{\mathcal{R}}_T(\cdot)$ is not used.*

Proof. On the retained coordinates, $\partial_n \bar{\mathcal{R}}_T(0;n) = -\sum_{j\le m} a_j^2\,\tilde{\sigma}_{r,j}^2/\big(n^2(a_j+\mu)^2\big) < 0$ (the truncation index m is locally constant in n almost everywhere), and $\bar{\mathcal{R}}_T(0;n) \downarrow \bar{\mathcal{R}}_T(0;\infty)$ as $n \to \infty$. The map $\kappa \mapsto \bar{\mathcal{R}}_T(0; n_r + \kappa n_h)$ is thus a continuous strict bijection from $[0,\infty)$ onto $(\bar{\mathcal{R}}_T(0;\infty),\, \bar{\mathcal{R}}_T(0;n_r)]$, which contains $m_\star$ in case (i) and not in case (ii). □

Proposition 6 (the fixed-μ cap is not permanent missing coverage)**.** *Suppose Σ_r is full rank and the robot-only baseline is allowed to retune its ridge penalty $\mu_n \to 0$ as $n \to \infty$. Then its risk $\bar{\mathcal{R}}_{T,\mu_n}(0;n) \to 0$. Consequently, whenever the fixed-μ mixture optimum $m_\star$ is strictly positive, a finite exchange solution exists for this tuned baseline; $\kappa_{\mathrm{ex}} = +\infty$ is possible only when $m_\star = 0$. Thus the fixed-μ cap in Proposition 5(ii) does not by itself establish coverage that no amount of robot data can reach.*

Proof. At $\mu = 0$ the robot-only per-coordinate risk is $\tilde{\sigma}_{r,j}^2/(a_j n)$ before the Σ_r weighting, so

$$\bar{\mathcal{R}}_{T,0}(0;n) = \frac{1}{n}\sum_{j\le m} \tilde{\sigma}_{r,j}^2 + \sum_{j>m} a_j \beta_j^{*2}.$$

Both terms tend to zero as $n \to \infty$ ($m \to p$), and the same limit holds for any $\mu_n \to 0$. For $m_\star > 0$, continuity and the zero limit give a finite n whose baseline risk equals $m_\star$. □

Finite rates greater than one: an example. At matched spectra $a = b = 1$ with $\beta^* = \delta = 0$, $\sigma_r^2 = 1$, and $\sigma_h^2 = 1/4$, inverse-variance weighting gives $m_\star = 1/((n_r + 4n_h)(1+\mu)^2)$, whereas $\bar{\mathcal{R}}_T(0;n) = 1/(n(1+\mu)^2)$. The defining equation yields the finite rate $\kappa_{\mathrm{ex}} = 4$: four target samples substitute for one source sample. This is distinct from the extended value $\kappa_{\mathrm{ex}} = +\infty$.

H.2. Per-coordinate unimodality, exactly

Drop the coordinate index and write the leading-order per-coordinate risk (effective variance plus bias square, including the mismatch-fluctuation term of Eq. (2)) as $\mathrm{MSE}(\lambda) = \mathrm{MSE}^{\mathrm{lead}}(\lambda) = N(\lambda)/D(\lambda)^2$ with

$$N(\lambda) = v_r(1-\lambda)^2 + v_h\lambda^2 + (u\lambda - w)^2 + \frac{2u^2}{n_h}\lambda^2, \tag{19}$$

and $D(\lambda) = D_0 + \Delta\lambda$, where $v_e = \tilde{\sigma}_e^2\lambda^e/n_e$ are the variance prices, $u = b\delta$, $w = \mu\beta^*$, $D_0 = a + \mu > 0$, $\Delta = b - a$, and $D > 0$ on $[0,1]$ since $D(1) = b + \mu > 0$. N is a quadratic, $N(\lambda) = n_0 + n_1\lambda + n_2\lambda^2$, with $n_0 = v_r + w^2$, $n_1 = -2(v_r + uw)$, $n_2 = v_r + v_h + u^2(1 + 2/n_h)$.

An affine identity for the derivative numerator is useful below.

Lemma 7. *$\partial_\lambda \mathrm{MSE}(\lambda) = L(\lambda)/D(\lambda)^3$ with L affine:*

$$L(\lambda) = p + q\lambda, \qquad p = n_1 D_0 - 2\Delta n_0, \qquad q = 2n_2 D_0 - \Delta n_1. \tag{20}$$

Proof. $\partial_\lambda \mathrm{MSE} = (N'D - 2\Delta N)/D^3$; in $N'D - 2\Delta N$ the quadratic terms $2n_2\Delta\lambda^2$ cancel exactly. □

Proposition 8 (coordinate risks are unimodal)**.** *L cannot cross zero from above on $[0,1]$: the pattern $L(0) > 0 > L(1)$ never occurs. Hence each coordinate risk is quasiconvex on $[0,1]$, with explicit minimizer*

$$\lambda^* = \begin{cases} 0, & L(0) \ge 0, \\ -p/q \in (0,1), & L(0) < 0 < L(1), \\ 1, & L(1) \le 0. \end{cases} \tag{21}$$

Proof. With $D_1 = D_0 + \Delta = b + \mu > 0$ and $s = uD_0 + \Delta w$, direct rearrangement gives the endpoint values

$$\begin{aligned} L(0) &= -2\big(v_r D_1 + ws\big), \\ L(1) &= 2\Big(\big(v_h + \tfrac{2u^2}{n_h}\big)D_0 + (u - w)\,s\Big). \end{aligned} \tag{22}$$

Suppose $L(0) > 0 > L(1)$. Since $v_r \ge 0$, $v_h + 2u^2/n_h \ge 0$, and $D_0, D_1 > 0$: if $s \ge 0$, the first inequality forces $w < 0$ and the second forces $u < w$; hence $s = uD_0 + \Delta w < w(D_0 + \Delta) = wD_1 < 0$, contradicting $s \ge 0$. If instead $s < 0$, they force $w > 0$ and $u > w$; hence $s > wD_1 > 0$, contradicting $s < 0$. The remaining sign patterns of the affine L give the classification (21); an interior zero of L is a sign change from negative to positive, hence a minimum. □

Remark 2 (sanity checks). At $\delta = 0$ and $\Delta = 0$ the interior branch of (21) reduces to $\lambda^* = v_r/(v_r + v_h)$, the classical inverse-variance weight. Moreover $L(0) = D_0^3\, \partial_\lambda \mathrm{MSE}(0^+)$, so the coordinate-wise marginal-harm condition $L(0) \geq 0$ is exactly the three-term decomposition of C1 of the main text.

H.3. Global unimodality: an exact region and a sufficient condition

Sums of quasiconvex functions need not be quasiconvex, so the aggregate $\bar{\mathcal{R}}_T(\lambda) = \sum_j a_j \,\mathrm{MSE}_j(\lambda)$ requires a separate argument.

Theorem 9 (matched spectra: quadratic regime)**.** *If $b_j = a_j$ for all j, then $\bar{\mathcal{R}}_T^{\mathrm{lead}}$ is a quadratic polynomial on $[0,1]$, $\bar{\mathcal{R}}_T^{\mathrm{lead}}(\lambda) = A_2\lambda^2 + A_1\lambda + A_0$, with*

$$A_2 = \sum_j \frac{a_j\, n_{2j}}{(a_j+\mu)^2}, \qquad A_1 = \sum_j \frac{a_j\, n_{1j}}{(a_j+\mu)^2}. \tag{23}$$

Whenever $A_2 > 0$ — i.e., unless the model is degenerate ($v_{rj} = v_{hj} = u_j = 0$ for every j) — $\bar{\mathcal{R}}_T$ is strictly convex, with the unique closed-form minimizer

$$\lambda^* = \mathrm{clip}_{[0,1]}\left(\frac{\sum_j a_j(v_{rj} + u_j w_j)/(a_j+\mu)^2}{\sum_j a_j\, n_{2j}/(a_j+\mu)^2}\right). \tag{24}$$

Proof. With $\Delta_j = 0$, $\mathrm{MSE}_j(\lambda) = N_j(\lambda)/(a_j+\mu)^2$; summing with weights a_j gives the quadratic, and the interior minimizer is $-A_1/(2A_2)$. □

Proposition 10 (convexity region)**.** *The per-coordinate second derivative is*

$$\mathrm{MSE}''(\lambda) = \frac{N''(\lambda)D(\lambda)^2 - 4\Delta\, N'(\lambda)D(\lambda) + 6\Delta^2 N(\lambda)}{D(\lambda)^4}, \tag{25}$$

and a sufficient condition for convexity of MSE_j on $[0,1]$ — hence, summing over j, for strict convexity and unimodality of $\bar{\mathcal{R}}_T$ — is

$$2\,|\Delta_j|\,\big(|n_{1j}| + 2n_{2j}\big)\, D_j^{\max} \;\leq\; n_{2j}\,\big(D_j^{\min}\big)^2 \qquad \text{for every } j, \tag{26}$$

where $D_j^{\max} = \max(a_j, b_j) + \mu$ and $D_j^{\min} = \min(a_j, b_j) + \mu$. Since $n_{2j} > 0$ and $D_j^{\min} > 0$, the condition holds whenever the per-coordinate spectral mismatch $|\Delta_j|/(a_j+\mu)$ is small enough, and is automatic at zero spectral gap.

Proof. Differentiate Lemma 7: $\mathrm{MSE}'' = (qD - 3\Delta L)/D^4$; substituting $q = N''D - \Delta N'$ and $L = N'D - 2\Delta N$ gives the displayed numerator. Since N is quadratic, $N'' = 2n_2$, $|N'| \leq |n_1| + 2n_2$ on $[0,1]$, $D \in [D^{\min}, D^{\max}]$, and $6\Delta^2 N \geq 0$; the numerator is therefore at least $2n_2(D^{\min})^2 - 4|\Delta|(|n_1| + 2n_2)D^{\max}$, which is nonnegative under (26). □

Outside the region (26) we have no proof of global unimodality, and positive combinations of quasiconvex rationals with distinct denominators can in principle be multimodal; however no multimodal instance was found in a randomized search over 4×10^3 aggregates (power-law and adversarial spectra), and $\bar{\mathcal{R}}_T(\lambda)$ is numerically unimodal on all grids used in this paper (Secs. G and I). In any case, Proposition 5 guarantees that κ_{ex} is well-defined irrespective of unimodality.

H.4. The source-sample axis: interior minimizers and an exact dosage law

Corollary 11 (interior minimizers from endpoint signs)**.** *If $\bar{\mathcal{R}}_T'(0^+) = \sum_j a_j L_j(0)/D_{0j}^3 < 0$ (the condition of C1 of the main text) and $\bar{\mathcal{R}}_T'(1^-) = \sum_j a_j L_j(1)/D_{1j}^3 > 0$, with $L_j(0), L_j(1)$ from Eq. (22) and $D_{1j} = b_j + \mu$, then neither $\lambda = 0$ nor $\lambda = 1$ is optimal and every global minimizer lies in $(0,1)$. No unimodality is needed for existence.*

Proof. $\bar{\mathcal{R}}_T'(0^+) < 0$ yields points with risk strictly below $\bar{\mathcal{R}}_T(0)$, and $\bar{\mathcal{R}}_T'(1^-) > 0$ yields points strictly below $\bar{\mathcal{R}}_T(1)$; a global minimizer exists by continuity on the compact interval. □

Lemma 12 (monotone value in n_h)**.** *At fixed λ, each coordinate risk is nonincreasing in n_h: both human-sample terms of N, $v_h\lambda^2 = \tilde{\sigma}_h^2 b\lambda^2/n_h$ and $2u^2\lambda^2/n_h$, have nonnegative coefficients. Hence the optimal gain $g(n_h) = \bar{\mathcal{R}}_T(0) - \min_\lambda \bar{\mathcal{R}}_T(\lambda; n_h)$ is nondecreasing: more human data never hurts at the optimal mixing ratio.*

Theorem 13 (dosage law for the leading-order risk; matched spectra)**.** *Assume $b_j = a_j$ for all j and $A_1 = \bar{\mathcal{R}}_T'(0^+) < 0$, and write $A_2(n_h) = A_2^r + B/n_h$ with*

$$A_2^r = \sum_j \frac{a_j(v_{rj} + u_j^2)}{(a_j+\mu)^2}, \qquad B = \sum_j \frac{a_j(\tilde{\sigma}_h^2 b_j + 2u_j^2)}{(a_j+\mu)^2}. \tag{27}$$

On the range where the minimizer is interior ($-A_1 < 2A_2(n_h)$):

(i) the optimal gain is hyperbolic, $g(n_h) = A_1^2/\big(4A_2(n_h)\big)$, with $g(0^+) = 0$, $g(\infty) = A_1^2/(4A_2^r)$, and half-saturation point $n_h^{1/2} = B/A_2^r$;

(ii) the worth-mixing threshold at tolerance $\varepsilon < g(\infty)$ is

$$g(n_h) \geq \varepsilon \iff n_h \geq n_h^{\varepsilon} = \frac{4\varepsilon B}{A_1^2 - 4\varepsilon A_2^r}; \tag{28}$$

(iii) the optimal ratio grows smoothly from zero, $\lambda^*(n_h) = -A_1/\big(2A_2(n_h)\big) \sim (-A_1/2B)\cdot n_h$ as $n_h \to 0$, approaching $\lambda_\infty^* = -A_1/(2A_2^r)$;

(iv) the exchange rate has the closed form

$$\kappa_{\mathrm{ex}}(n_h) = \frac{1}{n_h}\cdot\frac{n_r^2\, g(n_h)}{C - n_r\, g(n_h)}, \qquad C = \sum_j \frac{\tilde{\sigma}_{r,j}^2 a_j^2}{(a_j+\mu)^2}, \tag{29}$$

since at matched spectra $\bar{\mathcal{R}}_T(0;n) = C/n + \mathrm{const}$; the no-finite-solution regime of Proposition 5(ii) is the pole $g(n_h) \geq C/n_r$ ($\kappa_{\mathrm{ex}} = +\infty$). Note $C = K/2$ at zero spectral gap, with K the constant of the main-text H criterion.

Proof. By Theorem 9, $\bar{\mathcal{R}}_T(\lambda; n_h) = A_2(n_h)\lambda^2 + A_1\lambda + A_0$, and only A_2 depends on n_h. An interior minimizer gives $g = A_1^2/(4A_2)$ and $\lambda^* = -A_1/(2A_2)$; (i)–(iii) are rearrangements, with λ^* nondecreasing in n_h since A_2 decreases. For (iv), $\bar{\mathcal{R}}_T(0;n) = C/n + \sum_j a_j\mu^2\beta_j^{*2}/(a_j+\mu)^2$; solving $C/(n_r+\kappa_{\mathrm{ex}}n_h) = C/n_r - g(n_h)$ for κ_{ex} yields the formula, whose denominator vanishes exactly at $g(n_h) = C/n_r$ — the no-finite-solution condition $m_\star \leq \bar{\mathcal{R}}_T(0;\infty)$ of Proposition 5(ii). □

If $-A_1 \geq 2A_2(n_h)$ the minimizer sits at the boundary $\lambda^* = 1$ and $g(n_h) = -A_1 - A_2(n_h)$; since $A_2(n_h)$ decreases in n_h, the crossing happens at most once, at $n_h^1 = 2B/(-A_1 - 2A_2^r)$.

Remark 3 (the second transition is of a different kind). The two axes of the phase diagram differ in kind. Along n_r there is a genuine transition of the zero-mixing derivative (the main-text H criterion): the sign of $\bar{\mathcal{R}}_T'(0^+)$ flips at $H = 0$, and beyond n_r^{crit} adding a vanishing source fraction increases risk. This local statement alone does not prove that the global optimum is $\lambda^* = 0$; the stronger discard-all-source conclusion follows only where aggregate unimodality is proved (matched spectra or condition (26)), and is not asserted for general spectra. Along n_h there is no sign change: once $\bar{\mathcal{R}}_T'(0^+) < 0$, human data helps at *every* sample size, and the classical small-source threshold softens into the explicit n_h^{ε} of (ii) — equivalently, $\lambda^*(n_h) \leq \lambda_0$ for all $n_h \leq 2B\lambda_0/(-A_1 - 2A_2^r\lambda_0)$. The hard classical switch is recovered as the divergence $n_h^{\varepsilon} \to \infty$ when $A_1 \to 0^-$ or $B \to \infty$ (noisy or badly mismatched human data).

Global monotonicities of κ_{ex} do not follow merely from the zero-mixing decomposition in Eq. (15), because κ_{ex} depends on the optimized mixture value and an implicit baseline equation. The updated v12 numerics therefore report trends rather than theorem-level monotonicities: at zero spectral gap κ_{ex} ranges from 0.417 to 0.585 across the tested families; finite values greater than one occur (up to 6.866), while no-finite-solution cells first appear at spectral gaps between 0.5 and 1.0 depending on the family. Some grid cells are non-monotone in the spectral gap, and many large-gap cells have $\kappa_{\mathrm{ex}} = +\infty$; Spearman statistics on the remaining finite values are descriptive only. The fixed-μ cap can arise because the mixture optimum falls below the shrinkage floor $\bar{\mathcal{R}}_T(0;\infty)$; Proposition 6 shows that this does not establish permanent missing coverage when the baseline may retune μ. The capped display quantity $\kappa_{\mathrm{ex}}^{\mathrm{cap}} = \min(\kappa_{\mathrm{ex}}, 1)$ is not the exchange-rate definition.

I. Numerical verification protocol

The approximation level is stated for each protocol: the corrected risk $\bar{\mathcal{R}}_T^{\mathrm{det}}$ is used for Monte-Carlo accuracy curves, while the algebraic-identity checks for the marginal criterion, unimodality, exchange rate, and dosage law use $\bar{\mathcal{R}}_T^{\mathrm{lead}}$ as defined in Sec. E. Measured-estimator curves are not described as exact realizations of either deterministic approximation.

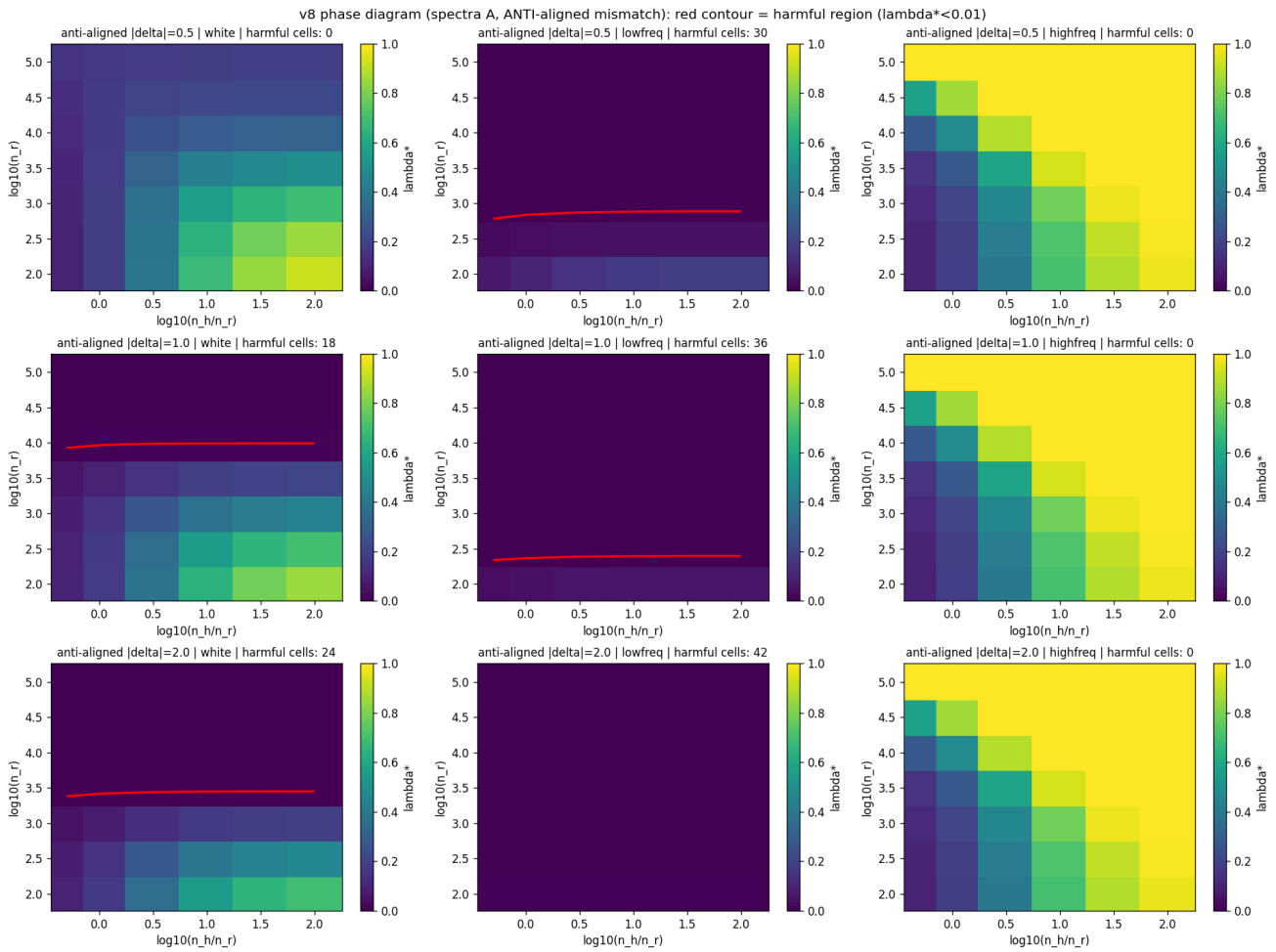


Figure 1. Optimal mixing ratio λ^* over the $(n_r, n_h/n_r)$ plane under anti-aligned mismatch (spectral family A), by mismatch magnitude (rows) and spectral location (columns). Red contour: harmful region ($\lambda^* < 0.01$). In this uncontrolled grid, low-frequency mismatch is harmful almost everywhere and high-frequency mismatch is harmless; location, target-weighted energy, and alignment vary together. The controlled candidate-family experiment in Sec. Q finds no sign flip of H after these invariants are fixed.

(v5) Accuracy of the deterministic equivalent. Three spectral/mismatch configurations $\times$ two mixing ratios, 200 Monte-Carlo replicates ($\sim$200k coordinate-replicates): binned median relative error 0.6–1.6%, aggregate and Σ_r-weighted aggregate error $\leq 2\%$. Ablations v3/v4 show the feedthrough term — not the ratio-fluctuation corrections — is the dominant closure: zero-order (population) formulas underestimate by 30–90%, ratio corrections alone do not close the gap, and feedthrough closes it to the stated accuracy.

(v6) Full ridge vs. diagonalized. Same-data comparison (50 replicates, $\lambda = 0.3$): the full ridge estimator is uniformly better (median full/diag risk ratio $0.64/0.50$ on spectral families A/B), *i.e.*, the feedthrough "noise" is partly recoverable signal for the full estimator; on the Σ_r-weighted risk our formula overestimates by 8% (A) / 28% (B). On these configurations the phase structure (sign conditions, transitions) is unchanged while quantitative thresholds shift; this motivates the full-ridge analysis below, not a proof of downstream transfer.

(v9) Falsification of a naive assembly. A leading-order power-law assembly predicts $n_r^{\text{crit}} \propto \mu^{-2.05}$; the measured exponent is -1.076. Diagnosis: the assembly drops the harm channel beyond the μ-induced truncation $j_\mu = \mu^{-1/\alpha_r}$ (where the cross term is amplified by μ^{-2}) and the mismatch-induced inflation of the feedthrough energy F_h. This failure is what led us to the H-functional form; we report it as the worked example of the falsification loop.

(v8) Harmful-region phase diagrams. Spectral family A, $n_r \in [10^2, 10^5]$, $n_h/n_r \in [0.5, 100]$: harmful-cell counts in Table 1 of the main text; the boundary expands monotonically in $\|\delta\|$ and toward small n_r.

(v12) Exchange-rate numerics as in Sec. H.

(v13) Unimodality verification. Randomized checks of the results of Sec. H (`verify_kex_unimodality.py`): the affine derivative identity of Lemma 7 matches finite differences to 7×10^{-6} (the finite-difference accuracy) over 2×10^5 random draws; the forbidden sign pattern of Proposition 8 never occurs, and all $\sim 10^5$ interior roots found are minima; the quadratic form of Theorem 9 holds to 2×10^{-14} with the closed-form λ^* matching the numerical argmin; condition (26) implies convexity in every one of the 1.8×10^4 draws in which it holds; and a targeted search over 4×10^3 aggregate risks (power-law and adversarial spectra) found no multimodal instance.

(v14) Dosage law and closed-form κ_{ex}. (`verify_dosage_law.py`.) Corollary 11: in 3.96×10^4 random general-spectrum draws with both endpoint signs strict, the global minimizer is interior in 100% of cases. Theorem 13: over 212 matched-spectrum configurations (8 decades of n_h each), the closed-form gain, the closed-form κ_{ex}, and the n_h^{ε} inversion match direct evaluation of the same leading-order per-coordinate risk (pointwise evaluation / bisection of the defining equation) to 3×10^{-10}, 1.1×10^{-10}, and 3×10^{-16} respectively, and $\lambda^*(n_h)$ is nondecreasing in every tested configuration.

(v16) Full-ridge numerical surrogate. (`verify_full_ridge.py`.) Monte-Carlo risk of the full ridge estimator against $\mathrm{MSE}^{\mathrm{full,num}}$: relative errors between -8.4% and $+0.4\%$ across four configurations, decaying with sample size ($-2.5\% \to +0.3\%$ as n_r grows $100 \to 6400$); on the same configurations the T1-a form overestimates by $+4\%$ to $+61\%$, consistent with the second-order feedthrough-recovery conjecture but not establishing a theorem-level upper bound; the mismatch sandwich converges to its per-coordinate price on these nondegenerate configurations (errors -0.3% to $+8\%$).

J. A label-shift floor for aligned representations (boundary)

This section formalizes boundary (a) stated in the main text: when alignment is enforced on representations whose *action marginals* differ, a shared deterministic head provably fails on the target domain. It is the squared-loss, mixture-distribution analogue of the classical bound of Zhao et al. (cited in the main text).

Setup: a representation $z = \Phi(x)$ with induced feature laws P_r^z, P_h^z satisfying $W_1(P_r^z, P_h^z) \le \varepsilon$; action marginals P_e^a with means μ_e, gap $\Delta\mu = \mu_h - \mu_r$; a deterministic L-Lipschitz head f; domain risks $R_e(f) = \mathbb{E}[(f(z) - y_e)^2]$.

Proposition 14 (label-shift floor)**.** *For every deterministic head f:*

(i) $R_r(f) + R_h(f) \ge \frac{1}{2}\big(|\Delta\mu| - L\varepsilon\big)_+^2$;

(ii) the mixture risk $R_\lambda(f) = (1-\lambda)R_r(f) + \lambda R_h(f)$ obeys

$$R_\lambda(f) \ge \lambda(1-\lambda)\big(|\Delta\mu| - L\varepsilon\big)_+^2; \tag{30}$$

(iii) any head with $R_\lambda(f) \le \eta$ pays target risk $R_r(f) \ge \big(|\Delta\mu| - L\varepsilon - \sqrt{\eta/\lambda}\big)_+^2$. We do not assert a stronger floor for an unspecified mixture-optimal head.

Proof. Write $\nu_e = \mathbb{E}_{z\sim P_e^z}[f(z)]$. By Jensen, $R_e(f) \ge (\nu_e - \mu_e)^2$, and by the Kantorovich–Rubinstein dual, $|\nu_r - \nu_h| \le L\, W_1(P_r^z, P_h^z) \le L\varepsilon$. (i) $(\nu_r-\mu_r)^2+(\nu_h-\mu_h)^2 \ge \frac{1}{2}\big((\nu_r-\mu_r)-(\nu_h-\mu_h)\big)^2 \ge \frac{1}{2}(|\Delta\mu|-L\varepsilon)_+^2$, since $(\nu_r-\mu_r)-(\nu_h-\mu_h) = -\Delta\mu + (\nu_r - \nu_h)$. (ii) Minimize $(1-\lambda)(\nu_r-\mu_r)^2 + \lambda(\nu_h-\mu_h)^2$ subject to $|\nu_r - \nu_h| \le L\varepsilon$. For fixed shift $\delta = \nu_h - \nu_r$, the minimum over the common level is attained at $\nu_r = (1-\lambda)\mu_r + \lambda(\mu_h - \delta)$ with value $\lambda(1-\lambda)(\Delta\mu - \delta)^2$; minimizing over $\delta \in [-L\varepsilon, L\varepsilon]$ gives $\delta^* = \mathrm{sign}(\Delta\mu)\min(L\varepsilon, |\Delta\mu|)$ and value $\lambda(1-\lambda)(|\Delta\mu| - L\varepsilon)_+^2$. (iii) $R_\lambda(f) \le \eta$ implies $|\nu_h - \mu_h| \le \sqrt{\eta/\lambda}$, hence $|\nu_r - \mu_r| \ge |\Delta\mu| - L\varepsilon - \sqrt{\eta/\lambda}$. Squaring and using $R_r(f) \ge (\nu_r - \mu_r)^2$ gives the displayed bound. □

Remark 4 (Bernoulli case). For binary actions $y \in \{0,1\}$, $|\Delta\mu| = D_{\mathrm{TV}}(P_r^a, P_h^a)$ and (i) is the Brier-score analogue of the classification bound of Zhao et al.: perfectly aligned representations with shifted label marginals are guaranteed to fail — exact alignment ($\varepsilon \to 0$) does not remove the floor, it creates it. For continuous actions the first-moment gap is the sharp quantity; second-moment (variance) shifts give analogous floors with second moments in place of means.

Remark 5 (operational reading for the L1 test). The floor is monotone in the marginal gap and applies to any head whose mixture risk is already small. It provides a target-risk bound, not an identification of the actual mixture-optimal head or of its exact risk. The quantitative prediction for the action-marginal-shift axis of the companion policy-level study is therefore stated only through the general low-mixture-risk bound, not through a separate λ^2 optimum formula. Numerical checks (`verify_zhao_floor.py`): bounds (i)–(iii) hold on 2×10^5 random draws with zero violations; the mean-level constrained quadratic solution matches brute-force minimization. These checks do not identify an optimal head in the original function class.

K. Theorem T1-b: a full-ridge population equivalent (fixed-p regime)

We prove a leading-order result for the fixed-dimension regime and explicitly separate it from second-order numerical corrections. The full ridge estimator is $\hat{\beta} = A^{-1}(w_r X_r^\top y_r + w_h X_h^\top y_h)$ with $A = w_r S_r + w_h S_h + \mu I$ and $S_e = X_e^\top X_e$ the (non-diagonal) sample Gram matrices, in the model of Sec. A. Define

$$\bar{\mathcal{R}}_T^{\text{full}}(\lambda) := \sum_j \frac{a_j}{D_j^2}\Big[V_j(\lambda) + (\lambda b_j \delta_j - \mu\beta_j^*)^2\Big],$$
$$V_j(\lambda) = \frac{(1-\lambda)^2\sigma_r^2 a_j}{n_r} + \frac{\lambda^2\sigma_h^2 b_j}{n_h}. \tag{31}$$

Theorem 15 (T1-b: population-level approximation)**.** *Let the covariates be Gaussian, p fixed, $\mu > 0$ fixed, and $n_r, n_h \to \infty$ at fixed $\lambda \in (0,1)$. Then*

$$\big|\mathcal{R}_T(\lambda) - \bar{\mathcal{R}}_T^{\text{full}}(\lambda)\big| = O(n^{-1/2}), \qquad n = \min(n_r, n_h), \tag{32}$$

where the constants depend on $(p, \mu^{-1}, \|\beta^*\|, \|\delta\|)$ and the spectra. The displayed approximation uses the true label-noise variances σ_e^2. Its resemblance to the diagonalized formula after replacing $\tilde{\sigma}_e^2$ is an algebraic observation, not a transfer of the diagonalized effective-noise mechanism to full ridge. This theorem does not *control $O(1/n)$ terms, derivatives with respect to λ, or errors of optimized risks.*

Proof. The error decomposes exactly as

$$\hat{\beta} - \beta^* = -\mu A^{-1}\beta^* + A^{-1} w_h S_h\, \delta + A^{-1}\eta,$$
$$\eta = w_r X_r^\top \varepsilon_r + w_h X_h^\top \varepsilon_h. \tag{33}$$

Two estimates drive everything: (1) Wishart concentration, $\|S_e/n_e - \Sigma_e\| = O_p(\sqrt{p/n_e})$, hence $\|A - D\| = O_p(n^{-1/2})$ with $D = (1-\lambda)\Sigma_r + \lambda\Sigma_h + \mu I = \operatorname{diag}(D_j)$; (2) resolvent perturbation, using the deterministic bound $\|A^{-1}\| \le 1/\mu$: on the event $\|A - D\| \le \mu/2$ (whose complement is exponentially unlikely),

$$\|A^{-1} - D^{-1}\| \;\le\; \|A - D\| \,/\, \big(\mu\,(\mu - \|A - D\|)\big) \;=\; O_p(n^{-1/2}). \tag{34}$$

For the mismatch term write $w_h S_h \delta = \lambda\Sigma_h\delta + \zeta$ with $\mathbb{E}\|\zeta\|^2 = (\lambda^2/n_h)\operatorname{Var}\big(x(x^\top\delta)\big) = O(1/n_h)$; hence the mismatch-plus-shrinkage risk converges to $\sum_j a_j(\lambda b_j\delta_j - \mu\beta_j^*)^2/D_j^2$, including the alignment cross term. For the variance, $\mathbb{E}[\eta\eta^\top \mid X] = w_r^2\sigma_r^2 S_r + w_h^2\sigma_h^2 S_h$, so $\mathbb{E}[\eta^\top A^{-1}\Sigma_r A^{-1}\eta]$ converges to $\operatorname{tr}\Sigma_r D^{-1}\big((1-\lambda)^2\sigma_r^2\Sigma_r/n_r + \lambda^2\sigma_h^2\Sigma_h/n_h\big)D^{-1}$, the displayed variance sum. Cross terms involving η vanish by conditional centering, and all replacements of A^{-1} by D^{-1} are justified by the deterministic bound $\|\Sigma_r^{1/2}A^{-1}\Sigma_e^{1/2}\| \le (\max_j a_j \cdot \max_j \lambda_j^e)^{1/2}/\mu^2$ on the good event. The complementary event is exponentially unlikely and the relevant estimators have finite fourth moments, so the stated order also holds for the expected risk $\mathcal{R}_T$; the remainder is not a stochastic symbol attached to an already-averaged object. □

Why population accuracy is not second-order accuracy. Take $p = 1$, $a = b = \beta^* = 1$, $\lambda = 1/2$, $\delta = 0.2$, $\mu = 0.1$, $n_r = n_h = 1000$, and zero label noise. Then $\lambda b\delta - \mu\beta^* = 0$ and $D = 1.1$, so Eq. (31) gives zero risk. The actual estimator nevertheless has positive risk from sample-Gram fluctuations: an independent 2×10^6-replicate calculation gives 1.6542×10^{-5} (standard error 1.66×10^{-8}), while the first nonzero fluctuation term is

$$\frac{2\lambda^2(b\delta)^2/n_h}{D^2} = 1.65289 \times 10^{-5}. \tag{35}$$

This is fully compatible with the $O(n^{-1/2})$ remainder, but demonstrates that the theorem cannot justify $O(1/n)$ coefficients or exact cancellation claims.

A legacy numerical surrogate, distinct from the proved T1-c object. Our earlier full-ridge numerical curves use $\mathrm{MSE}^{\text{full,num}}$: Eq. (31) with the mismatch-fluctuation term $2\lambda^2(b_j\delta_j)^2/(n_h D_j^2)$ and the same denominator corrections $-4N_{0j}\mathrm{Cov}(\tilde{n}_j, \hat{D}_j)/D_j^3 + 3N_{0j}^2\mathrm{Var}(\hat{D}_j)/D_j^4$ as in the diagonal analysis (`mse_t1b` in the code). This is a computational surrogate motivated by the diagonal expansion; it is not the second-order object proved in T1-c below. In the cancellation example above, the surrogate gives exactly Eq. (35), whereas the population display gives zero.

Consequences and non-consequences. (i) *Mismatch sandwich at a nondegenerate scale.* Write $\mathcal{B}^0_{\mathrm{mis}} = \sum_j a_j(\lambda b_j \delta_j / D_j)^2$. The same resolvent estimates give $\mathcal{B}_{\mathrm{mis}} = \mathcal{B}^0_{\mathrm{mis}}(1+o(1))$ when $\mathcal{B}^0_{\mathrm{mis}}$ dominates the $n^{-1/2}$ remainder scale. No relative-error claim is made when $\mathcal{B}^0_{\mathrm{mis}} = 0$ or is itself $O(1/n)$.

(ii) *Feedthrough recovery is a second-order conjecture with numerical support, not a consequence of T1-b.* The proposed difference $\sum_j a_j[(1-\lambda)^2 F_r a j/n_r + \lambda^2 F_h b_j/n_h]/D_j^2$ is $O(1/n)$, smaller than the theorem's remainder. Thus T1-b cannot establish its sign or magnitude, and we do not claim a theorem-level variance upper bound for T1-a; the v6/v16 results are numerical evidence only.

(iii) *No blanket downstream transfer.* Corollary 1, Theorem 2, Corollary 2, and the dosage law involve derivatives, $O(1/n)$ coefficients, or optimized risks. They therefore do not follow from T1-b by substituting $\tilde{\sigma}_e^2 \to \sigma_e^2$. The complete fixed-$p$ second-order risk and uniform derivatives are proved separately below without assumptions on the number or location of population optima. Their consequences for optimized risk are then stated under the respective interior, boundary, flat, or multiple-optimum conditions; these do not follow from T1-b alone.

K.1. Complete second-order risk at fixed dimension

Theorem 16 (T1-c: full-ridge second-order risk)**.** *Under the Gaussian fixed-p assumptions of Theorem 15, let $\rho = n_h/n_r$ be fixed. Then, uniformly for $\lambda \in [0,1]$,*

$$R_{\mathrm{full}}(\lambda; n_r) = R_0(\lambda) + \frac{R_1(\lambda)}{n_r} + o(n_r^{-1}). \tag{36}$$

Define $D = (1-\lambda)\Sigma_r + \lambda\Sigma_h + \mu I$, $R = D^{-1}$, $S = R\Sigma_r R$, $v = -\mu\beta^* + \lambda\Sigma_h\delta$, and

$$\begin{aligned} \mathcal{T}_e(N) &= \Sigma_e N \Sigma_e + \mathrm{tr}(\Sigma_e N)\Sigma_e, \\ \mathcal{T} &= (1-\lambda)^2\mathcal{T}_r + \lambda^2\rho^{-1}\mathcal{T}_h, \\ U &= \mathcal{T}(R), \\ C &= RUS + SUR + R\mathcal{T}(S)R. \end{aligned} \tag{37}$$

The coefficients are $R_0 = v^\top S v$ and

$$\begin{aligned} R_1 = v^\top C v &+ \frac{\lambda^2}{\rho}\,\mathrm{tr}\{S\mathcal{T}_h(\delta\delta^\top)\} \\ &- \frac{2\lambda^2}{\rho} v^\top[R\mathcal{T}_h(S) + S\mathcal{T}_h(R)]\delta \\ &+ (1-\lambda)^2\sigma_r^2\,\mathrm{tr}(S\Sigma_r) + \frac{\lambda^2\sigma_h^2}{\rho}\,\mathrm{tr}(S\Sigma_h). \end{aligned} \tag{38}$$

This is a risk expansion, not yet a uniform derivative or optimized-risk theorem.

Proof. Write $n = n_r$, $w = \lambda$, $G_e = X_e^\top X_e/n_e$, $Z_e = G_e - \Sigma_e$, $Z = (1-w)Z_r + wZ_h$, and $\xi = wZ_h\delta$. For $t \in [0,1]$, define $G_{e,t} = \Sigma_e + tZ_e$ and

$$\begin{aligned} B_t &= [(1-w)G_{r,t} + wG_{h,t} + \mu I]^{-1}, \\ q_t &= v + t\xi, \qquad W_t = (1-w)^2\sigma_r^2 G_{r,t} + w^2\rho^{-1}\sigma_h^2 G_{h,t}, \\ F_0(w,t) &= q_t^\top B_t \Sigma_r B_t q_t, \qquad F_1(w,t) = \mathrm{tr}(\Sigma_r B_t W_t B_t). \end{aligned} \tag{39}$$

Conditional centering and covariance of the label noise give the exact identity

$$R_{\mathrm{full}}(w; n) = \mathbb{E}F_0(w,1) + n^{-1}\mathbb{E}F_1(w,1). \tag{40}$$

The interpolation matrices $G_{e,t} = (1-t)\Sigma_e + tG_e$ are positive semidefinite, so $\|B_t\| \le \mu^{-1}$ for every design and all real $w, t \in [0,1]$. Lemma 18 below bounds the expected Taylor remainders uniformly: expanding F_0 to order two in t costs $O(n^{-3/2})$, and expanding $n^{-1}F_1$ to order one costs $O(n^{-2})$. That lemma depends only on Gaussian Gram moments and this deterministic resolvent bound, not on the present expansion.

We identify the Taylor coefficients. At $t = 0$, $B_0 = R$, and the coefficients through degree two in $B_t \Sigma_r B_t$ are

$$\begin{aligned} S, \qquad & H_1 = -RZS - SZR, \\ & H_2 = RZRZS + SZRZR + RZSZR. \end{aligned} \tag{41}$$

Consequently the bias coefficients through degree two are

$$v^\top S v, \qquad 2\xi^\top S v + v^\top H_1 v, \qquad v^\top H_2 v + 2v^\top H_1 \xi + \xi^\top S \xi. \tag{42}$$

All linear terms have zero expectation. For any deterministic symmetric matrix M, independence of the two designs and the real Wishart identity give

$$\mathbb{E}[Z_e M Z_e] = \mathcal{T}_e(M)/n_e, \quad \mathbb{E}[ZMZ] = \mathcal{T}(M)/n, \quad \mathbb{E}[ZMZ_h] = \frac{w}{\rho n}\mathcal{T}_h(M). \tag{43}$$

It follows that $\mathbb{E}H_2 = C/n$ and

$$\begin{aligned} 2\mathbb{E}[v^\top H_1 \xi] &= -\frac{2w^2}{\rho n} v^\top [R\mathcal{T}_h(S) + S\mathcal{T}_h(R)]\delta, \\ \mathbb{E}[\xi^\top S \xi] &= \frac{w^2}{\rho n} \operatorname{tr}\{S\mathcal{T}_h(\delta\delta^\top)\}. \end{aligned} \tag{44}$$

For the noise term, $F_1(w, 0)$ equals the last two terms of R_1 in Eq. (38), and $\mathbb{E}\partial_t F_1(w, 0) = 0$, since this derivative is linear in the centered Gram fluctuations. Combining these identities with the uniform expected remainder bounds proves T1-c, with an $O(n^{-3/2})$ remainder, hence the stated $o(n^{-1})$. □

Numerical verification of T1-c. The script `verify_full_ridge_second_order.py` checks 22 design-level conditional-risk cells over $p = 1, 2, 20$, aligned and anti-aligned mismatch, $\lambda \in \{0, 0.1, 0.3, 0.5, 0.8, 1\}$, and $n_r = 1000, 4000$. All simulated means are within 2.13 Monte-Carlo standard errors of Eq. (36); three direct actual-estimator checks are within 1.12 standard errors. Compact-versus-pairwise algebra and consistency with the zero-mixing derivative have maximum error 7.11×10^{-15}. The one-dimensional cancellation example recovers the first positive risk term. The legacy surrogate is compared separately in `compare_full_ridge_surrogate.py`; it is not the object proved here.

K.2. The standard full-ridge zero-mixing derivative

Theorem 17 (Fixed-p derivative with the first Gram correction)**.** *Under the Gaussian fixed-p assumptions of Theorem 15, the actual full-ridge target risk satisfies*

$$R'_{\text{full}}(0^+; n_r) = H_{\text{full}} - \frac{K_{\text{full}}}{n_r} + o(n_r^{-1}). \tag{45}$$

The limit in the shared spectral basis is

$$H_{\text{full}} = -2\mu \sum_j \frac{a_j b_j \beta_j^* \delta_j}{(a_j + \mu)^2} - 2\mu^2 \sum_j \frac{a_j \beta_j^{*2}(b_j - a_j)}{(a_j + \mu)^3}, \tag{46}$$

but K_{full} is not obtained by differentiating the population risk alone. Write $\Sigma = \Sigma_r$, $D_0 = \Sigma + \mu I$, $R = D_0^{-1}$, $S = R\Sigma R$, $M = \Sigma_h - \Sigma$, and

$$\mathcal{T}(N) = \Sigma N \Sigma + \operatorname{tr}(\Sigma N)\Sigma, \qquad U = \mathcal{T}(R). \tag{47}$$

Define the coefficient matrices

$$\begin{aligned} \mathcal{C} &= RUS + SUR + R\mathcal{T}(S)R, \\ \mathcal{L} &= \mathcal{J}(M) + R\mathcal{T}(S)R + 2SUR, \\ \mathcal{J}(N) &= RUSNR + SURNR + SNRUR \\ &\quad + R\mathcal{T}(S)RNR + R\mathcal{T}(SNR)R + S\mathcal{T}(RNR)R. \end{aligned} \tag{48}$$

Then

$$K_{\text{full}} = K_0 - H_1, \qquad K_0 = 2\sigma_r^2 \operatorname{tr}\{\Sigma(R\Sigma R + R\Sigma RMR)\},$$
$$H_1 = -2\mu(\Sigma_h\delta)^\top \mathcal{C}\beta^* - 2\mu^2\beta^{*\top}\mathcal{L}\beta^*. \tag{49}$$

The expansion is uniform in a sufficiently small interval $\lambda \in [0, \varepsilon]$.

Proof sketch. Let $G = X_r^\top X_r/n_r$, $B = (G + \mu I)^{-1}$, and $Z = G - \Sigma$. Differentiating the exact error decomposition, averaging over the independent human design and label noise, and using $\mathbb{E}[\eta_0\eta_0^\top \mid X_r] = \sigma_r^2 G/n_r$ gives

$$\begin{aligned} R'_{\text{full}}(0^+) = & -2\mu(\Sigma_h\delta)^\top \mathbb{E}[B\Sigma B]\beta^* \\ & - 2\mu^2\beta^{*\top}\mathbb{E}[B\Sigma B(\Sigma_h - G)B]\beta^* \\ & - \frac{2\sigma_r^2}{n_r}\operatorname{tr}\Sigma\,\mathbb{E}[BGB] \\ & - \frac{2\sigma_r^2}{n_r}\operatorname{tr}\Sigma\,\mathbb{E}[B(\Sigma_h - G)BGB]. \end{aligned} \tag{50}$$

For deterministic N, the real Wishart moment is $\mathbb{E}[ZNZ] = \mathcal{T}(N)/n_r$. On the event $\|Z\| \le \mu/2$, expand $B = R - RZR + RZRZR + O_p(\|Z\|^3)$. Pairing the two Z factors yields

$$\begin{aligned} \mathbb{E}[B\Sigma B] &= S + \mathcal{C}/n_r + o(n_r^{-1}), \\ \mathbb{E}[B\Sigma B(\Sigma_h - G)B] &= SMR + \mathcal{L}/n_r + o(n_r^{-1}). \end{aligned} \tag{51}$$

The last two terms in Eq. (50) contribute $-K_0/n_r + O(n_r^{-2})$; collecting the bias correction H_1/n_r proves Eq. (49). The complementary event is exponentially unlikely at fixed p, while $\|A^{-1}\| \le \mu^{-1}$ and the finite-moment assumptions control its contribution. The same bounds permit differentiation under the expectation and are uniform near $\lambda = 0$. □

Remark 6 (Identifiability and the critical-size interpretation). Set $t = 1/n_r$ and $\bar{g}(t) = H_{\text{full}} - K_{\text{full}}t$. A positive asymptotic critical scale $n_r^{\text{crit,full}} = K_{\text{full}}/H_{\text{full}}$ requires $H_{\text{full}}K_{\text{full}} > 0$, and then $\bar{g}'(t^*) = -K_{\text{full}} \neq 0$. The sign is identifiable away from the root at any fixed relative distance. At the root itself the leading term vanishes and the remainder is still $o(n_r^{-1})$, so this is not an exact finite-sample root theorem. In particular, if $H_{\text{full}} > 0$ but $K_{\text{full}} \le 0$, no positive critical size follows from this expansion. This result does not transfer the dosage law, exchange rate, or global optimum.

Numerical verification of the full-ridge derivative. The script `verify_full_ridge_zero_derivative.py` checks the compact matrices against an independent pairwise resolvent expansion (maximum discrepancy 1.11×10^{-16}) and against exact design-level conditional derivatives generated by Bartlett Wishart sampling. Across $p = 1, 2, 20$ and $n_r = 1000$–16000, all six Monte-Carlo cells agree with the prediction within sampling and higher-order error. For spectral family A at $p = 20$, $n_r = 1000$ gives prediction 0.0148136 and Monte Carlo 0.0147617 ± 0.0000197; at $n_r = 16000$ the values are 0.0263331 and 0.0263323 ± 0.0000052. Direct actual-estimator finite differences at $\lambda = 10^{-4}$ give 0.0143334 ± 0.0003422 for the former cell. These checks concern the zero-mixing derivative, not the general-λ numerical surrogate.

Remark 7 (the proportional regime). For $p \asymp n$ the resolvent perturbation step fails and an anisotropic deterministic equivalent is needed. The structure of a prospective proof is visible in the leave-one-out expansion of the sandwich: writing $S_h = \sum_i x_i x_i^\top$, the cross pairs $i \neq k$ carry the leading order through $\mathbb{E}[x\,(x^\top\delta)] = \Sigma_h\delta \neq 0$, while the diagonal pairs $i = k$ contribute only $O(1/n_h)$ corrections; a proportional-regime proof must therefore control $\sum_{i\neq k}(\delta^\top x_i)(x_i^\top A^{-1}\Sigma_r A^{-1}x_k)(\delta^\top x_k)$ via leave-two-out resolvent expansions — the route through Wu's pipeline flagged in the project notes. The fixed-p second-order expansion and the nondegenerate interior-optimum theorem below do not extend to this regime; boundary-switching and general global-shape control also remain separate tasks.

L. Uniform derivatives and nondegenerate interior optima

Lemma 18 (Uniform interpolation and expected remainder bounds)**.** *Fix p, $\mu > 0$, $0 < \rho < \infty$, both covariance matrices, the task vectors, and the noise variances. With the interpolation and functions F_0, F_1 in Eq. (39), put $z = \|Z_r\| + \|Z_h\|$. For every fixed $q > 0$,*

$$\mathbb{E}z^q = O(n^{-q/2}), \qquad n = n_r. \tag{52}$$

For $k = 0, 1, 2$, $j = 0, 1, 2, 3$, and $\ell = 0, 1$, there are deterministic constants, independent of n, the design realization, w, and t, such that

$$\sup_{w,t\in[0,1]} |\partial_w^k \partial_t^j F_\ell(w,t)| \le C_{k,j} z^j (1+z)^{k+2}. \tag{53}$$

For $j = 0$ the factor z^0 is one. In particular, expectation and up to two w derivatives can be interchanged, including one-sided endpoint limits.

Proof. Each entry of Z_e is an average of n_e independent centered products of Gaussian coordinates. In an even moment of order $2m$, a nonzero expectation requires each sample index to appear at least twice. There are $O(n_e^m)$ such index tuples, and their moments are bounded by fixed Gaussian moments. Hence each entry has $2m$-th absolute moment $O(n_e^{-m})$. Finite dimension, the bound of operator norm by the sum of absolute entries, and the moment inequality for smaller powers give Eq. (52). Here $n_h = \rho n$; all constants may depend on the fixed parameters and on the moment order.

For the derivative bound, write $A_t = B_t^{-1}$. The deterministic positive-semidefinite interpolation gives $A_t \succeq \mu I$. Moreover,

$$\begin{aligned} \|\partial_w A_t\| &= \|\Sigma_h - \Sigma_r + t(Z_h - Z_r)\| \le C(1+z), \\ \|\partial_t A_t\| &= \|(1-w)Z_r + wZ_h\| \le z, \qquad \|\partial_w \partial_t A_t\| \le z. \end{aligned} \tag{54}$$

All other nonzero derivatives of A_t are covered by these displays. Repeatedly using $\partial B_t = -B_t(\partial A_t)B_t$ and the product rule shows

$$\sup_{w,t\in[0,1]} \|\partial_w^k \partial_t^j B_t\| \le C_{k,j} z^j (1+z)^k. \tag{55}$$

Indeed each t derivative contributes a factor bounded by z, each w derivative contributes at most a factor $C(1+z)$, and every resolvent factor is bounded by μ^{-1}. Mixed derivatives of A_t satisfy the same counting bound.

The vectors q_t and their w derivatives have norm at most $C(1+z)$; their nonzero t derivatives have norm at most Cz. The same bounds apply to W_t and its first two w derivatives, with every nonzero t derivative bounded by Cz. Applying the product rule to the two quadratic/trace expressions in Eq. (39) proves Eq. (53). The two q_t factors account for the exponent $k+2$; this exponent also bounds the noise expression, which has only one W_t factor.

These bounds are integrable by Eq. (52). Dominated differentiation on $(0,1)$ and dominated limits at the endpoints justify the asserted exchange with expectation. No event truncation or complex extension is needed. □

Theorem 19 (Uniform risk derivatives)**.** *Under the fixed-p assumptions of Theorem 16, for $k = 0, 1, 2$ and uniformly for $\lambda \in [0, 1]$,*

$$R_{\text{full}}^{(k)}(\lambda; n_r) = R_0^{(k)}(\lambda) + R_1^{(k)}(\lambda)/n_r + o(n_r^{-1}). \tag{56}$$

Endpoints use one-sided derivatives.

Proof. Fix $k \in \{0, 1, 2\}$ and apply Taylor's formula in the real interpolation parameter t to $\partial_w^k F_0(w,t)$ and $\partial_w^k F_1(w,t)$. All mixed derivatives exist by positive definiteness. Define

$$\begin{aligned} T_0(w) &= F_0(w,0) + \partial_t F_0(w,0) + \tfrac{1}{2}\partial_t^2 F_0(w,0), \\ T_1(w) &= F_1(w,0) + \partial_t F_1(w,0). \end{aligned} \tag{57}$$

The integral remainder formula and Lemma 18 give

$$\begin{aligned} \mathbb{E} \sup_{w\in[0,1]} \left|\partial_w^k\{F_0(w,1) - T_0(w)\}\right| &\le C_k \mathbb{E}[z^3(1+z)^4] = O(n^{-3/2}), \\ n^{-1}\mathbb{E} \sup_{w\in[0,1]} \left|\partial_w^k\{F_1(w,1) - T_1(w)\}\right| &\le C_k n^{-1} \mathbb{E}[z^2(1+z)^4] = O(n^{-2}). \end{aligned} \tag{58}$$

For example, the bias remainder is $\frac{1}{2}\int_0^1 (1-t)^2 \partial_t^3 \partial_w^k F_0(w,t)\, dt$. The noise remainder uses $\int_0^1 (1-t)\partial_t^2 \partial_w^k F_1(w,t)\, dt$. Expanding $(1+z)^4$ and applying Eq. (52) proves the stated expectation orders directly.

The moment calculations in the proof of T1-c identify the exact deterministic function

$$\mathbb{E}T_0(w) + n^{-1}\mathbb{E}T_1(w) = R_0(w) + n^{-1}R_1(w). \tag{59}$$

The lemma permits differentiating this identity and Eq. (40) under the expectation. Therefore

$$\sup_{w\in[0,1]}\left|R_{\mathrm{full}}^{(k)}(w;n)-R_0^{(k)}(w)-n^{-1}R_1^{(k)}(w)\right|=O(n^{-3/2})=o(n^{-1}). \tag{60}$$

The same uniform bounds cover the one-sided derivatives at 0 and 1. In particular, derivative control follows from expected mixed-derivative remainder bounds, not from differentiating an unspecified real-axis $o(n^{-1})$ term. The proof uses only the label-noise second moments after conditioning on the designs; the assumed fourth moments are more than sufficient. □

Theorem 20 (Nondegenerate interior optimum)**.** *Suppose R_0 has a unique global minimizer $w_0\in(0,1)$ with $R_0''(w_0)>0$. Write $R_n(w)=R_{\mathrm{full}}(w;n_r)$. For all sufficiently large n_r, R_n has a unique global minimizer w_n^* on $[0,1]$, and*

$$\begin{aligned} w_n^* &= w_0-\frac{R_1'(w_0)}{n_rR_0''(w_0)}+o(n_r^{-1}),\\ R_n(w_n^*) &= R_0(w_0)+R_1(w_0)/n_r+o(n_r^{-1}),\\ \mathcal{G}_n:=R_n(0)-R_n(w_n^*) &= \mathcal{G}_0+\mathcal{G}_1/n_r+o(n_r^{-1}), \end{aligned} \tag{61}$$

where $\mathcal{G}_0=R_0(0)-R_0(w_0)$ and $\mathcal{G}_1=R_1(0)-R_1(w_0)$. When $R_1'(w_0)=0$, the display gives $n_r(w_n^*-w_0)\to 0$; the direction of the shift is then determined only at higher order.

Proof. Choose a neighborhood I_δ of w_0 on which $R_0''>0$. Uniform second-derivative control makes R_n strictly convex there. Outside I_δ, the unique population minimizer leaves a positive risk gap, and uniform risk control excludes a global minimizer. The signs of R_n' at the two endpoints of I_δ follow from uniform first-derivative control, so the unique interior convex minimizer is global. Since δ is arbitrary, $w_n^*\to w_0$. The optimized-risk display then follows by continuity and the uniform risk expansion.

It remains to prove the weight rate. Put $c=R_0''(w_0)>0$, $d=R_1'(w_0)$, and $A_*=-d/c$. The explicit matrix coefficients in Theorem 16 are analytic in a fixed neighborhood of w_0. For every fixed A, the uniform first-derivative expansion evaluated at w_0+A/n_r, Taylor's theorem applied to R_0', and continuity of R_1' give

$$\begin{aligned} n_rR_n'(w_0+A/n_r) &= n_rR_0'(w_0+A/n_r)+R_1'(w_0+A/n_r)+o(1)\\ &= cA+d+o(1). \end{aligned} \tag{62}$$

For any $A_-<A_*<A_+$, the limits $cA_-+d<0$ and $cA_++d>0$ imply that, for all sufficiently large n_r, R_n' is negative at w_0+A_-/n_r and positive at w_0+A_+/n_r. These points eventually lie in I_δ, where R_n is strictly convex, so the stationary point between them is the unique global minimizer w_n^*. Choosing $A_-=A_*-\varepsilon$ and $A_+=A_*+\varepsilon$ for arbitrary $\varepsilon>0$ proves $n_r(w_n^*-w_0)\to A_*$, as claimed. □

Theorem 21 (Nondegenerate and flat left-boundary optima)**.** *Suppose $w=0$ is the unique global minimizer of R_0 on $[0,1]$. If $R_0'(0^+)>0$, then $w_n^*=0$ for all sufficiently large n_r. If instead $R_0'(0^+)=0$ and $c=R_0''(0^+)>0$, then the sign of $d=R_1'(0^+)$ determines the direction: $d>0$ gives $w_n^*=0$ for all sufficiently large n_r, while $d<0$ gives*

$$w_n^*\sim -d/(cn_r),\qquad \mathcal{G}_n\sim d^2/(2cn_r^2). \tag{63}$$

The case $d=0$ is undetermined at this order. The right boundary is analogous when $R_0'(1^-)<0$.

Proof sketch. In the nondegenerate case, uniform first-derivative control preserves strict increase near zero, while the population risk gap and uniform risk control exclude all other global minimizers. In the flat case, $d>0$ gives the same boundary conclusion. For $d<0$, evaluating $R_n'(A/n_r)=(cA+d+o(1))/n_r$ on both sides of $A=|d|/c$ proves $n_rw_n^*\to|d|/c$. For the gain, integrate the derivative expansion rather than subtracting the $o(n_r^{-1})$ risk remainders: $\mathcal{G}_n=-\int_0^{w_n^*}R_n'(w)\,dw$. The integral has the required $o(n_r^{-2})$ remainder and yields $\mathcal{G}_n\sim d^2/(2cn_r^2)$. □

A local-harm/global-benefit counterexample. The script `verify_full_ridge_counterexample.py` verifies a three-dimensional configuration with $R_0'(0^+)=0.2628138>0$ but an interior global minimizer $w_0=0.0682897$ satisfying $R_0(0)=0.0338052$ and $R_0(w_0)=0.0247415$, with $R_0''(w_0)=3.44659$. At $n_r=1000$, the actual zero-mixing derivative is 0.221174 (prediction 0.221356), while the optimized gain is 0.009396 (prediction 0.009405). Hence $R_0'(0^+)>0$ alone cannot imply global discard; the global-minimum-at-zero shape condition in Theorem 21 is essential. Boundary scaling checks use `verify_full_ridge_boundary.py`, including a high-replication flat-boundary cell.

Numerical verification. The script `verify_full_ridge_global_optimum.py` checks 16 design-level derivative cells and two paired actual-estimator center differences, followed by nondegenerate optima in the one-dimensional cancellation example and a two-dimensional aligned example. Direct derivative checks are within 0.73 standard errors; four optimized-risk checks are within 1.59 standard errors. In the latter example, $w_0 = 0.3708597$ and $R_0''(w_0) = 0.13996$; at $n_r = 1000, 4000$, the predicted optimized risks are 0.01063196 and 0.01002290, compared with Monte Carlo 0.01063452 and 0.01002470.

The separate script `verify_full_ridge_weight_expansion.py` checks the new weight rate by fitting roots of average conditional-derivative curves, rather than averaging per-replicate argminima. In the two-dimensional example, $R_1'(w_0) = 0.320971$ and the limiting scaled shift is -2.29329. The optima of the first-order coefficient curve have scaled shifts $-2.07343, -2.17785, -2.23408, -2.26330$ at $n_r = 500, 1000, 2000, 4000$, approaching this limit. The $n_r = 500$ cell has visible higher-order deviation and is not counted as an asymptotic fit success. At $n_r = 4000$, 200,000 conditional-design replicates give a Monte Carlo scaled shift of -2.29688 ± 0.07803. In the near-zero cancellation example, the predicted constant is -0.09091; all eight 20,000-replicate root estimates are within 1.98 Monte Carlo standard errors, although these intervals are comparatively wide. Independent actual-E2 paired derivative checks at $n_r = 1000$ give -0.000435 ± 0.000137 and 0.000469 ± 0.000137 on the left and right of the first-order weight prediction, with the expected signs. These checks do not address boundary-switching or flat/multiple optima.

M. Global shape of full-ridge population risk

With $D_j(w) = (1-w)a_j + wb_j + \mu$, write

$$R_0(w) = \sum_j a_j \frac{(wb_j\delta_j - \mu\beta_j^*)^2}{D_j(w)^2}. \tag{64}$$

M.1. Log-convexity at zero mismatch

Proposition 22 (Zero-mismatch classification)**.** *Let $\delta = 0$ and $\mathcal{J} = \{j : a_j\beta_j^{*2} > 0\}$. Then R_0 is log-convex. It is strictly log-convex with a unique minimizer whenever some $j \in \mathcal{J}$ has $a_j \neq b_j$. If every active coordinate has $a_j = b_j$, or $\mathcal{J}$ is empty, then R_0 is constant. In the strict case, the signs of $R_0'(0^+)$ and $R_0'(1^-)$ classify whether the unique optimum is at zero, one, or an interior point.*

Proof. For $j \in \mathcal{J}$,

$$\frac{d^2}{dw^2}\log D_j(w)^{-2} = \frac{2(b_j - a_j)^2}{D_j(w)^2} \geq 0. \tag{65}$$

Thus D_j^{-2} is log-convex, strictly when $a_j \neq b_j$. Positive weighted sums preserve log-convexity and strictness if any summand is strict. Strict log-convexity implies strict convexity and hence at most one minimizer; continuity gives existence. When all active denominators are constant, R_0 is constant. The endpoint classification follows from monotonicity of the derivative of a strictly convex function. □

Proposition 23 (One coordinate cannot have multiple valleys)**.** *For $p = 1$, $R_0'(w) = L(w)/D(w)^3$ with affine L. Hence R_0 has at most one nondegenerate interior local valley; if $L \equiv 0$, it is constant.*

Proof. An affine nonzero L has at most one root, so the derivative can change sign at most once. Multiple valleys would require alternating signs with at least three roots. □

M.2. Polynomial global certification

For each coordinate, $R_{0j}'(w) = L_j(w)/D_j(w)^3$ with affine L_j. Consequently,

$$R_0'(w) = \sum_j a_j \frac{L_j(w)}{D_j(w)^3}, \qquad P(w) = \sum_j a_j L_j(w) \prod_{k\neq j} D_k(w)^3, \tag{66}$$

where P and R_0' have the same zeros on $[0,1]$ and $\deg P \leq 3p-2$. For fixed parameters, complete real-root isolation, multiplicity checks, and comparison of all stationary and boundary candidates certify the global minimizers. A nonconstant derivative has at most $3p-2$ stationary roots; a derivative that vanishes identically is treated as a separate flat case.

M.3. A two-dimensional multimodal certificate

Take $p = 2$, $\mu = 7/1000$, $(a_1, a_2) = (1, 1/2)$, and $(b_1, b_2) = (1, 1/16)$, corresponding to exact power laws $\alpha_r = 1$ and $\alpha_h = 4$. Define $(c_1, c_2) = (1/50, 19/20)$ and $(u_1, u_2) = (\sqrt{26}/1000, \sqrt{3}/500)$. Set $\beta_j^* = u_j c_j/\mu$ and $\delta_j = u_j/b_j$. Then

$$\beta^* = (\sqrt{26}/350, 19\sqrt{3}/70), \qquad \delta = (\sqrt{26}/1000, 4\sqrt{3}/125), \tag{67}$$

and $\sum_j j^2 \beta_j^{*2} = 54163/61250 < 1$, so the example satisfies the task ellipsoid with $s = 1$. With $K_j = a_j u_j^2$,

$$R_0(w) = \sum_{j=1}^{2} K_j \frac{(w - c_j)^2}{D_j(w)^2}, \quad (K_1, K_2) = (13/500000, 3/500000), \tag{68}$$

where $D_1 = 1007/1000$ and $D_2 = 507/1000 - 7w/16$.

After clearing denominators, the sign of R_0' is determined by the negative of

$$\begin{aligned} Q(w) = {} & 870898437500w^4 - 3045158593750w^3 \\ & + 3569273662500w^2 - 1647923289900w + 238369257759. \end{aligned} \tag{69}$$

All four roots are simple and lie respectively in $(0, 1/2)$, $(2/3, 3/4)$, $(3/4, 1)$, and $(1, 2)$. Thus exactly three stationary points lie inside $(0, 1)$:

$$w_1 \approx 0.2648422643, \quad w_2 \approx 0.7270105700, \quad w_3 \approx 0.8691124964. \tag{70}$$

Since $R_0'(0) < 0$, they are a local minimum, a local maximum, and a local minimum. Rational interval evaluation gives

$$\begin{aligned} R_0(w_1) &\in [1.9948444989, 1.9948445856] \times 10^{-5}, \\ R_0(w_2) &\in [2.1174449516, 2.1174451514] \times 10^{-5}, \\ R_0(w_3) &\in [2.0929101838, 2.0929103414] \times 10^{-5}. \end{aligned} \tag{71}$$

Moreover $R_0(0) = 2.1076278346 \times 10^{-5}$ and $R_0(1) = 2.7729880412 \times 10^{-5}$. These intervals prove that w_1 is the unique global minimizer. Together with Proposition 23, this proves that $p = 2$ is the minimum dimension for multiple valleys.

M.4. Valley bifurcations and two global minimizers

Define

$$R_t(w) = K_1 \frac{(w - c_1)^2}{D_1(w)^2} + tK_2 \frac{(w - c_2)^2}{D_2(w)^2}, \qquad t > 0. \tag{72}$$

The discriminant of its stationary polynomial has two positive zeros

$$\begin{aligned} t_1 &= \frac{38034733763}{35580951312} - \frac{1257113\sqrt{384090}}{5930158552} \approx 0.9375848094, \\ t_2 &= \frac{38034733763}{35580951312} + \frac{1257113\sqrt{384090}}{5930158552} \approx 1.2003419385. \end{aligned} \tag{73}$$

The leading coefficient does not vanish, and $P(t, 0)$ and $P(t, 1)$ have no nonnegative roots. Sturm root counts therefore give the complete partition: one simple interior root for $0 < t < t_1$; one simple and one double root at t_1; three simple roots for $t_1 < t < t_2$; one double and one simple root at t_2; and one simple root for $t > t_2$.

At $t = 1$, the first branch is the unique global minimizer; at $t = 11/10$, the third branch is the unique global minimizer. To prove that two global minimizers occur, take the bracket $T_* = [1.06085925, 1.06085926]$. It contains three simple continuous root branches. The first-minus-third branch-risk difference is negative at the left endpoint and positive at the right endpoint, while interval evaluation shows that the middle stationary point and both boundaries remain strictly above the two local minima throughout T_*. By the intermediate value theorem, some $t_* \in T_*$ has two global minimizers. Independent numerical evaluation gives $t_* \approx 1.0608592591022064$, with minimizers 0.2955060494 and 0.8833510935 and common risk approximately $2.1057410481 \times 10^{-5}$.

The scripts `verify_global_risk_shape.py` and `verify_global_risk_shape_p2_family.py` reproduce these root isolations, discriminant checks, interval certificates, and independent numerical comparisons. The results concern R_0, not R_1 or R_n.

N. Flat and multiple population optima

N.1. Exact R_1 form at matched spectra and zero mismatch

Let $\Sigma_r = \Sigma_h = \Sigma$ and $\delta = 0$. Define $R = (\Sigma + \mu I)^{-1}$, $S = R\Sigma R$, and

$$\begin{aligned} \mathcal{T}(N) &= \Sigma N \Sigma + \operatorname{tr}(\Sigma N)\Sigma, \\ \mathcal{C} &= R\mathcal{T}(R)S + S\mathcal{T}(R)R + R\mathcal{T}(S)R, \qquad \Gamma = \mu^2 \beta^{*\top} \mathcal{C} \beta^*. \end{aligned} \tag{74}$$

With $N_r = \sigma_r^2 \operatorname{tr}(S\Sigma)$ and $N_h = \sigma_h^2 \operatorname{tr}(S\Sigma)$, define

$$A = N_r + \Gamma, \qquad B = (N_h + \Gamma)/\rho. \tag{75}$$

Direct simplification of T1-c yields

$$R_0(w) = C, \qquad R_1(w) = A(1-w)^2 + Bw^2. \tag{76}$$

To verify nonnegativity, write $r_j = (a_j + \mu)^{-1}$, $s_j = a_j r_j^2$, $q = \operatorname{tr}(\Sigma R)$, and $z = \operatorname{tr}(\Sigma S)$. In the common eigenbasis,

$$\mathcal{C}_{jj} = 3a_j^3 r_j^4 + 2a_j q\, r_j s_j + a_j z\, r_j^2 \geq 0. \tag{77}$$

Thus $\Gamma \geq 0$, and $A, B \geq 0$. Equality $A + B = 0$ occurs exactly when $\beta^* = 0$ and $\sigma_r^2 = \sigma_h^2 = 0$.

Theorem 24 (Optimal weight, risk, and gain at flat R_0)**.** *If $A + B > 0$, then R_1 has the unique minimizer $w_1^* = A/(A+B)$. For sufficiently large n_r, R_n is strictly convex on $[0,1]$, $w_n^* \to w_1^*$, and*

$$\begin{aligned} R_n(w_n^*) &= C + \frac{AB}{A+B}\frac{1}{n_r} + o(n_r^{-1}), \\ \mathcal{G}_n &= \frac{A^2}{A+B}\frac{1}{n_r} + o(n_r^{-1}). \end{aligned} \tag{78}$$

If $A = B = 0$, then $R_n(w) \equiv 0$ for every sample size. Every weight is optimal and $\mathcal{G}_n = 0$ exactly.

Proof. When $A + B > 0$, $R_1'' = 2(A+B) > 0$ everywhere, so the uniform second-derivative expansion makes R_n strictly convex for sufficiently large n_r. The rescaled objectives $n_r(R_n - C)$ converge uniformly to R_1, whose unique minimizer has a positive value gap outside any fixed neighborhood. Hence $w_n^* \to w_1^*$. The optimized-risk formula follows from this convergence and the uniform expansion. Finally, $\mathcal{G}_n = R_n(0) - R_n(w_n^*)$ and $R_1(0) = A$ give the gain formula. If $A = B = 0$, positivity of the covariance eigenvalues and μ in Eq. (76) and its coefficient definitions forces $\beta^* = 0$ and $\sigma_r^2 = \sigma_h^2 = 0$. Since $\delta = 0$ and the noises have zero mean, both labels vanish almost surely. Full ridge then returns zero for every design and weight, proving the exact zero-risk statement. □

N.2. R_1 branch selection at multiple population optima

Proposition 25 (First-order branch selection)**.** *Suppose R_0 has exactly two global minimizers $0 < w_1 < w_2 < 1$, both with positive local curvature. If $R_1(w_1) < R_1(w_2)$, then for all sufficiently large n_r the global optimum occurs only near w_1, and $w_n^* \to w_1$. The gap between the two local branch minima is $(R_1(w_2) - R_1(w_1))/n_r + o(n_r^{-1})$. If $R_1(w_1) = R_1(w_2)$, first-order analysis does not select a branch; higher-order terms or exact symmetry may determine whether a tie persists.*

Proof. Choose disjoint closed interior neighborhoods of w_1, w_2 on which $R_0'' > 0$, with R_0' negative at each left endpoint and positive at each right endpoint. Uniform first- and second-derivative control preserves these signs and strict convexity for large n_r, giving a unique local minimizer $w_{i,n}$ in each neighborhood. The local argument of Theorem 20 gives $w_{i,n} = w_i - R_1'(w_i)/(n_r R_0''(w_i)) + o(n_r^{-1})$. Thus $R_n(w_{i,n}) = R_0^* + R_1(w_i)/n_r + o(n_r^{-1})$. Outside these neighborhoods, the population risk has a strictly positive gap above R_0^*, and uniform risk control excludes global minimizers there. Comparing the two branch minima proves the result. Equality of the first-order coefficients leaves the comparison unresolved. □

Fixed-candidate risk comparison, separate from branch selection. For the two-global-minimum example above, with $\sigma_r^2 = 1, \sigma_h^2 = 4, \rho = 2$, the values at the two fixed population weights are $R_1(w_1) = 1.5697698402$ and $R_1(w_2) = 5.1438849070$. At $n_r = 1000$, the first-order prediction for $R_n(w_2) - R_n(w_1)$ is 0.0035741151, and 20,000 conditional-risk replicates give $0.0035763806 \pm 0.0000007785$ (mean $\pm$ one standard error). This experiment evaluates those fixed weights, not the finite-sample local minimizers $w_{i,n}$; it supports their risk ordering but does not validate branch persistence or the optimized branch-gap expansion. The discrepancy from the first-order fixed-point prediction is about 2.91 Monte Carlo standard errors; higher-order approximation error is distinct from sampling uncertainty.

For the first population minimizer, $R_0''(w_1) \approx 2.37736 \times 10^{-5}$ and $R_1'(w_1) \approx -0.330613$, giving a formal displacement of approximately 13.91 at $n_r = 1000$. Its size rules out interpreting this cell as a small-displacement validation. The theorem is asymptotic; an experimental validation of branch selection must first establish that the two finite-sample local minima persist and then compare their optimized risks.

The script `verify_flat_r0_r1.py` checks 200 random matched-spectrum configurations, with maximum quadratic-identity error 1.82×10^{-12} and maximum flat-R_0 error 3.55×10^{-15}. Four conditional-risk gain checks are within approximately 1.18 Monte Carlo standard errors. The script `verify_r1_rn_global_shapes.py` reproduces the fixed-candidate risk comparison and additional general-R_1 shape examples that are not needed for the present claims.

O. Further consequences: matched-spectrum gain, exchange, and surrogate dosage

These extensions use the risk expansions developed above. Their distinct approximation scopes are stated separately; they are not required for the main text's global-certification and selection results.

For matched spectra, $R_0(w) = A_0 + A_1 w + A_2 w^2$ with A_2 independent of $\rho = n_h/n_r$, and $R_1(w;\rho) = C_0(w) + C_1(w)/\rho$. If $A_2 > 0$, the population optimum is $w_0 = \mathrm{clip}_{[0,1]}(-A_1/(2A_2))$, with

$$G_0 = \begin{cases} 0, & A_1 \geq 0, \\ A_1^2/(4A_2), & A_1 < 0,\ -A_1 < 2A_2, \\ -A_1 - A_2, & A_1 < 0,\ -A_1 \geq 2A_2. \end{cases} \tag{79}$$

Under the nondegenerate optimal-branch assumptions, $\mathcal{G}_n = G_0 + G_1(\rho)/n_r + o(n_r^{-1})$, where $G_1 = R_1(0) - R_1(w_0) = D_0 + D_1/\rho$. Equivalently, $\mathcal{G}_n = G_0 + D_0/n_r + D_1/n_h + o(n_r^{-1})$; this is a fixed-$\rho$ expansion, not a claim of uniformity as $\rho \to 0$ or ∞.

Exact dosage rule for a quadratic surrogate. Write $S = R\Sigma R$, $C_r = \sigma_r^2 \operatorname{tr}(S\Sigma)$, and $C_h = \sigma_h^2 \operatorname{tr}(S\Sigma)$. Retaining population label noise defines the quadratic surrogate

$$\widetilde{R}_{n_r,n_h}(w) = R_0(w) + \frac{C_r(1-w)^2}{n_r} + \frac{C_h w^2}{n_h}. \tag{80}$$

Its linear and quadratic coefficients are

$$\begin{aligned} A_1^{\mathrm{lead}} &= A_1 - 2C_r/n_r, \\ A_2^{\mathrm{lead}}(n_h) &= A_2^r + B/n_h, \\ A_2^r &= A_2 + C_r/n_r, \qquad B = C_h. \end{aligned} \tag{81}$$

This surrogate omits sample-Gram fluctuation corrections that can be of order n_r^{-1}, the same order as its retained noise terms. Its dosage identity is therefore not a complete first-order full-ridge dosage expansion. For $A_2^{\mathrm{lead}} > 0$, a nonnegative A_1^{lead} gives zero optimal gain at $w = 0$. When $A_1^{\mathrm{lead}} < 0$, the hyperbolic gain $(A_1^{\mathrm{lead}})^2/(4A_2^{\mathrm{lead}})$ applies only when $-A_1^{\mathrm{lead}} < 2A_2^{\mathrm{lead}}(n_h)$; otherwise the boundary gain is $-A_1^{\mathrm{lead}} - A_2^{\mathrm{lead}}(n_h)$. If $-A_1^{\mathrm{lead}} > 2A_2^r$, the boundary is reached at $n_h^b = B/(-A_1^{\mathrm{lead}}/2 - A_2^r)$, and the boundary saturation value is $-A_1^{\mathrm{lead}} - A_2^r$ rather than the interior hyperbolic limit.

First-order exchange approximation and its pole. The target-only risk is $S + B_0/m + o(m^{-1})$, with $S = R_0(0)$ and $B_0 = R_1(0)$, while the optimized mixture risk is $S - G_0 + (B_0 - G_1)/n_r + o(n_r^{-1})$. Assume $B_0 > 0$ and define $\Gamma_n^{(1)} = n_r G_0 + G_1$. Dropping the remainders gives the approximate exchange equation $B_0[1/n_r - 1/(n_r + \kappa n_h)] = \Gamma_n^{(1)}/n_r$.

For $0 < \Gamma_n^{(1)} < B_0$, its finite positive solution is

$$\kappa_{\mathrm{ex}}^{(1)} = \frac{\Gamma_n^{(1)}}{\rho(B_0 - \Gamma_n^{(1)})}. \tag{82}$$

When $\Gamma_n^{(1)} = 0$, the solution is zero; for $\Gamma_n^{(1)} \geq B_0$, the approximate equation has no finite solution. If $G_0 > 0$, the pole is eventually crossed at fixed ρ. These are algebraic statements about the first-order equation and the fixed-μ baseline. They do not classify the actual finite-sample exchange rate near the pole, where discarded terms may change the sign. No uniform exchange error bound or unrestricted source-size dosage guarantee is claimed.

Numerical verification. The script `verify_full_ridge_dosage_exchange.py` compares the closed-form optimum with grid minimization of the same quadratic surrogate in 16 dose cells, with maximum gain discrepancy 1.49×10^{-10}, including a surrogate boundary crossing at $n_h = 1779.1732$. This is an algebraic implementation check. Separately, for one full-ridge interior configuration, $G_1(\rho) = 0.99757865 - 1.203415/\rho$; at $n_r = 1000, \rho = 6$, the predicted gain is 0.0122185831 and 50,000 conditional-risk replicates give $0.0122262407 \pm 0.0000066100$. The first-order exchange equation has no finite solution for the same configuration at $n_r = 1000$ in the tested ratios, while an algebraic $n_r = 40, \rho = 6$ example has $\kappa_{\mathrm{ex}}^{(1)} = 2.8527963$. General spectra, tuned regularization, and higher-order exchange corrections are not asserted.

P. Pre-registered S5 direct-sample and global validation

The pre-registered S5 suite fixes n_h/n_r and scans n_r directly. Primary computations use exact label-noise-conditional risks with Bartlett Wishart sampling; independent checks use actual labels and the standard E2 estimator. Main boundary cells use 400 replicates and 1000 bootstrap resamples; the two root tests use Bonferroni 97.5% intervals. The preregistration, raw JSON, seeds, and failure cells are provided in the repository directory `s5_global`.

Table 2. Direct full-ridge root tests; both pass the 20% compatibility criterion. The entire ratio interval must lie in $[0.8, 1.2]$.

Configuration	Root (predicted / estimated)	97.5% ratio CI
p2 anti-aligned	55.644 / 61.750	[1.082, 1.137]
p20 spectral A	453.395 / 464.610	[1.008, 1.041]

Actual-E2 paired center differences have the same sign on both sides of each root. For the local-harm/global-benefit example, the design-level derivative at $n_r = 1000$ is 0.22652 while the optimal gain is 0.00936; actual-E2 paired gain is 0.009266 ± 0.000372. The matched-interior actual-E2 paired gain is 0.011610 ± 0.000437, compatible with its first-order prediction 0.0122186.

The 16 matched-dose cells compare the analytical optimum with grid minima of the same quadratic surrogate, with maximum gain discrepancy 1.49×10^{-10} and a surrogate crossing to $w = 1$ at $n_h = 1779.17$. They verify the implementation of the surrogate formula. Exchange checks concern the first-order equation, including no-finite-solution cells at $n_r = 1000$ and a finite algebraic example $\kappa_{\mathrm{ex}}^{(1)} = 2.8527963$ at $n_r = 40, \rho = 6$.

Flat-boundary cells preserve the original 400-replicate results and add pre-registered noise diagnostics. At $n_r = 10^5$, theory gives $w = 0.084335$ and gain 1.361×10^{-6}; high-repetition Monte Carlo gives $w = 0.073090$ and gain $(1.214 \pm 0.159) \times 10^{-6}$. At $n_r = 10^6$, the values are $w = 0.008433$ and 1.361×10^{-8} versus $w = 0.007871$ and $(1.238 \pm 0.173) \times 10^{-8}$. These are compatible with the asymptotic rates. The $n_r = 10^4$ cell remains nonlocal and is not reported as a uniform-fit success.

All S5 evidence is controlled-model evidence. It does not address nonlinear policies, real visual features, tuned regularization, or arbitrary non-unimodal classification.

Q. Experimental details for the trained-model validation

Generator. Gaussian covariates with power-law spectra in a shared (rotatable) eigenbasis; $\beta_j^* \propto j^{-(s+1/2)}$ with $s = 1$ normalized to the ellipsoid $\sum_j j^{2s} \beta_j^{*2} = 1$; $\sigma_r^2 = 1$, $\sigma_h^2 = 4$; $p = 1000$; $n_r = 1000$, $n_h = 6000$ unless stated; $\mu = 0.05$; spectral family A $(\alpha_r, \alpha_h) = (2.0, 0.9)$, family B $(1.2, 1.0)$; mismatch shapes low-frequency j^{-1}, white j^0, high-frequency $(j/p)^2$, negated for the anti-aligned regime ($c < 0$). Because β^* is known exactly, target-domain excess risk is computed without proxies.

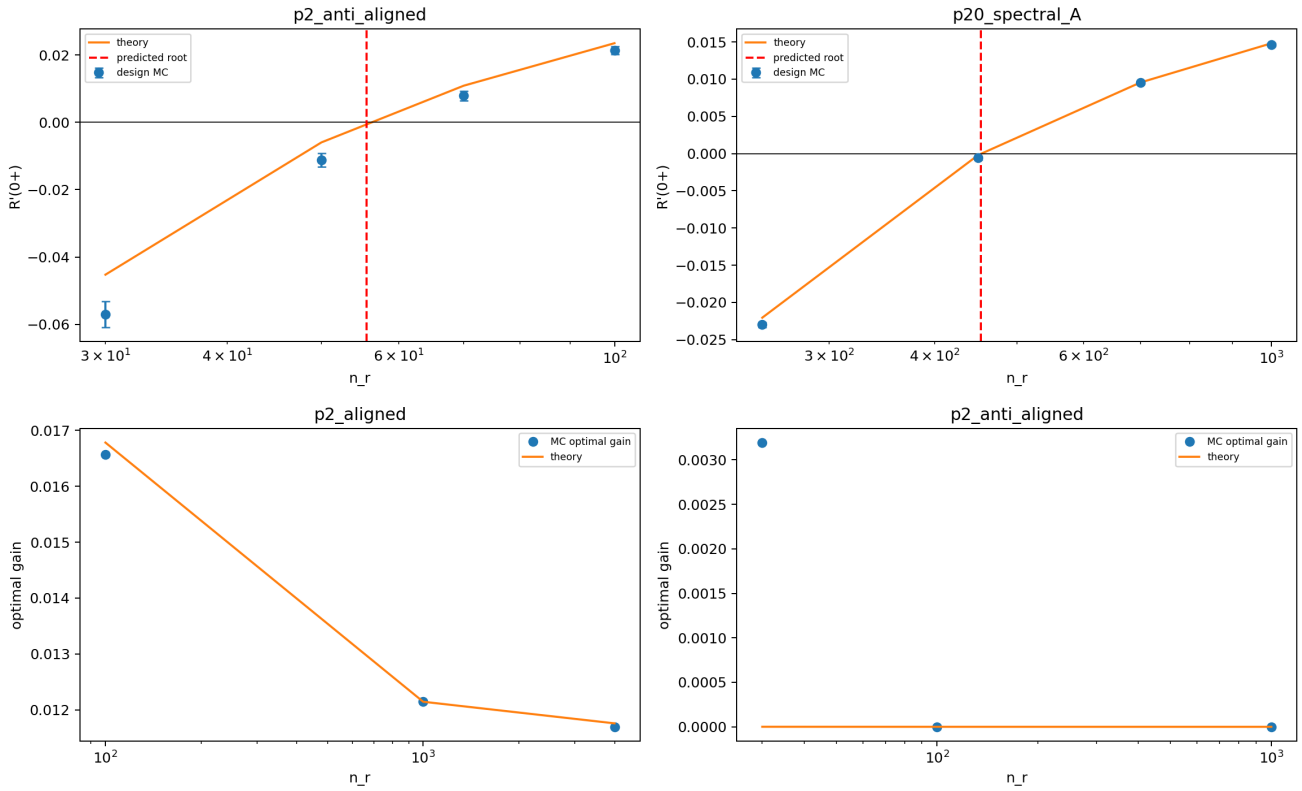


Figure 2. Pre-registered direct-n_r derivative and optimal-gain checks. Raw draws and bootstrap procedures are summarized in `summary.json`.

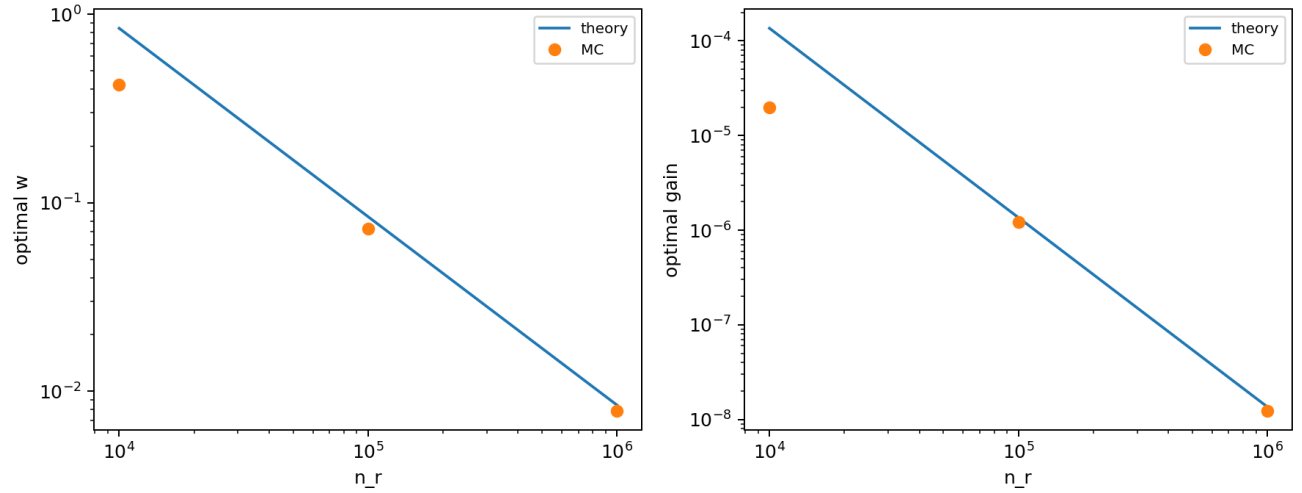


Figure 3. Flat-boundary optimal weight and gain on logarithmic axes. High-repetition diagnostics reduce argmin noise without changing the preregistered criterion.

Estimator ladder. E1: diagonal ridge, closed form (200 replicates). E2: full ridge, closed form with per-sample mixture weights (200 replicates). E3: linear model trained by full-batch Adam ($\mathrm{lr} = 3 \times 10^{-3}$, 20k steps, cosine decay, zero init), loss $\sum_i w_i (y_i - x_i^\top \beta)^2 + \mathrm{l2}\, \|\beta\|^2$ so that $\mu_{\mathrm{eff}} = \mathrm{l2}$ by construction ($\mathrm{l2} = 0.05$; N1e uses optimizer-level weight decay instead);

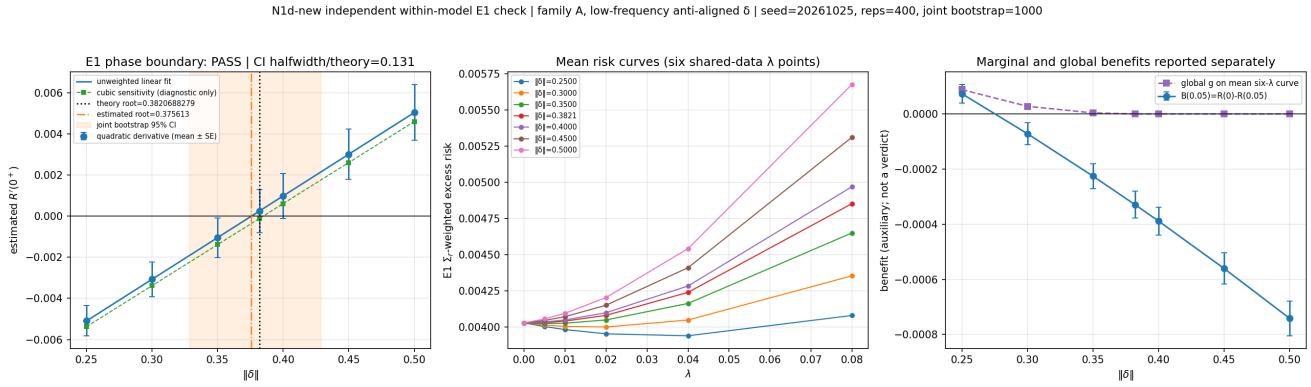


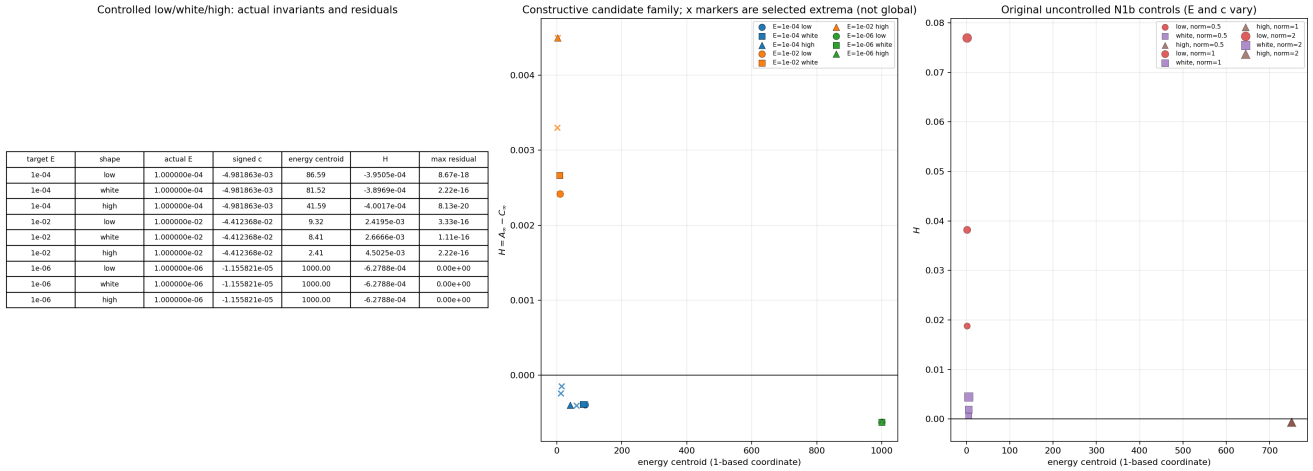
Figure 4. **Independent within-model E1 phase-boundary check.** Left: direct estimates of $R'(0^+)$ versus mismatch norm, with the theoretical root inside the joint-bootstrap 95% CI. Center: near-zero mean risk curves. Right: fixed-λ and global gains are reported separately and do not enter the verdict. This is same-model compatibility evidence, not external or equivalence validation.

single seed in this coarse pass. E4: two-layer MLP (width 512, ReLU), Adam lr $= 10^{-3}$, batch 512, 5k steps, explicit L_2 on weight matrices; risk on a 50k-sample Monte-Carlo test set; single seed.

Per-experiment outcomes (coarse pass). N1a (bridge): E1 vs. formula 0.150%, E2 vs. the numerical surrogate $\mathrm{MSE}^{\mathrm{full,num}}$ 0.766%, full-batch-Adam E3 vs. E2 10^{-15} (median, an optimizer–objective identity check), E4 ∼32% (single-seed report only). N1b (uncontrolled location contrast): sign agreement 9/9 (E2), 8/9 (E3), 9/9 (E4, exploratory single seed); the low/high-frequency contrast is retained as an uncontrolled observation, not as independent location evidence. N1c (dosage): hyperbolic fit wins the initial comparison ($R^2 = 0.99701$ vs. 0.99346 exp, 0.98502 log) after extending n_h to $\approx 8\times$ the half-saturation point; fitted half-saturation 3331 vs. theory 3865; the pre-registered strict $1/g$-linearity gate **fails** ($R^2 = 0.92571 < 0.995$), so exact dosage linearity is unconfirmed. N1d (H boundary): global truncated-gain sign agreement with the finite-n_r criterion 26/27 for E2 and E3 (not a direct zero-derivative test); point-boundary gates **fail** (E2 0.30, E3 0.75 vs. marginal root 0.382), while the sign-change bracket is $[0.25, 0.30]$; benefit magnitude $\sim 2\times10^{-5}$ is below the $\sim 5\times10^{-5}$ noise floor. N1e (scalar ridge inversion): in the two tested linear configurations, L_2-coupled Adam gives $\mu_{\mathrm{eff}} = \mathrm{wd}/2$ at the two measured nonzero wd values; decoupled AdamW gives small inverted coefficients ($\mu_{\mathrm{eff}} \approx 2.5\times10^{-7}$–$3.8\times10^{-7}$ across $\mathrm{wd} \in [0, 10^{-2}]$), with residuals $\sim 10^{-5}$ in configuration A and 8.24% in configuration B. This is a fitting diagnostic, not evidence that AdamW provides essentially no regularization; sensitivity to learning rate, training duration, parameter scale, and seeds is untested. N1f (basis rotation): for E2, median formula error grows from 0.96% to 30.5% over $\theta = 0° \to 90°$ (doubling at 15°), with no benefit-sign flip in the tested grid; E3 already has a sign flip at $\theta = 0°$, so no sign-robustness claim is made for E3.

Independent within-model E1 checks (pre-registered, fresh seeds). N1d-new directly estimates $R'(0^+)$ from six near-zero mixing ratios on E1 (400 replicates), rather than substituting a global truncated-gain sign. The derivative-versus-mismatch root is 0.3756 with joint-bootstrap 95% CI $[0.3286, 0.4288]$, containing the finite-n_r theoretical root 0.3821; the CI half-width is 13.1% of the theory value, so the pre-registered compatibility gate is **PASS**. CI inclusion is not an equivalence test or external validation. N1c-new uses 400 fresh replicates at each of 11 human sample sizes and a zero-parameter E1 dosage curve with per-coordinate feedthrough. The measured gains are compatible with that curve ($\chi^2 = 11.124$, df $= 11$, $p = 0.433$), meaning no significant departure is detected rather than equivalence being proved; the half-saturation estimate is 3541 (95% CI $[2948, 4243]$) versus 3624 theoretically, and the saturation gain is 0.003724 (CI $[0.003491, 0.003992]$) versus 0.003662. Shapes fit only for $n_h \le 4000$ give held-out RMSE 3.51×10^{-4} (hyperbola), 3.83×10^{-4} (logarithmic), and 8.64×10^{-4} (exponential); this is a limited three-point shape comparison, not external validation. These checks concern E1 and the leading-order risk; they do not establish migration to E2/E3.

Controlled mismatch experiment (constructive candidate family). With $\|\delta\|$, $\delta^\top \Sigma_r \delta$, and c fixed, 21 controlled directions have maximum constraint residual 3.33×10^{-16}. At $E = 10^{-4}$, all directions have $H < 0$ (range $[-4.052 \times 10^{-4}, -1.517\times10^{-4}]$); at $E = 10^{-2}$, all have $H > 0$ (range $[2.420\times10^{-3}, 4.502\times10^{-3}]$), corresponding to $n_r^{crit} = 3154$–5870. At $E = 10^{-6}$ the feasible set is a single point, so no location comparison is possible. No sign flip occurs in any nondegenerate configuration: the supported conclusion is *amplitude variation, not an independent location-driven phase transition*. The extrema are selected within the disclosed constructive candidate family, not certified as global extrema. E1



target E	shape	actual E	signed c	energy centroid	H	max residual
1e-04	low	1.000000e-04	-4.981863e-03	86.59	-3.9505e-04	8.67e-18
1e-04	white	1.000000e-04	-4.981863e-03	81.52	-3.8969e-04	2.22e-16
1e-04	high	1.000000e-04	-4.981863e-03	41.59	-4.0017e-04	8.13e-20
1e-02	low	1.000000e-02	-4.412368e-02	9.32	2.4195e-03	3.33e-16
1e-02	white	1.000000e-02	-4.412368e-02	8.41	2.6666e-03	1.11e-16
1e-02	high	1.000000e-02	-4.412368e-02	2.41	4.5025e-03	2.22e-16
1e-06	low	1.000000e-06	-1.155821e-05	1000.00	-6.2788e-04	0.00e+00
1e-06	white	1.000000e-06	-1.155821e-05	1000.00	-6.2788e-04	0.00e+00
1e-06	high	1.000000e-06	-1.155821e-05	1000.00	-6.2788e-04	0.00e+00

Figure 5. **Controlled mismatch comparison.** Left: actual invariants and residuals for the controlled low/white/high directions. Center: H versus energy centroid within the constructive candidate family; no sign flip. Right: the original uncontrolled comparison in which energy and alignment also vary. The candidate-family extrema are not global-extremum claims.

independently checks three representative directions at $E = 10^{-4}$ (400 fresh replicates): derivative means are -0.01396, -0.01378, and -0.01373, with all bootstrap 95% CIs below zero, consistent with $H < 0$ and the finite-n_r correction. No sample-size bracketing for a quantitative n_r^{crit} claim is performed.

Q.1. CIFAR real-feature protocol

Features and domains. CIFAR-10 Python batches are decoded in channel-first layout $(N, 3, 32, 32)$. Grayscale images are the unweighted mean over the RGB channel axis. A 32-component PCA basis and feature mean are fitted on the 50,000 CIFAR training images only; the official test batch is reserved for final evaluation. Target samples come from classes 0–4 and auxiliary samples from classes 5–9. The original class labels are used only to define these domains, not as regression targets.

Controlled labels. The target regression vector has ten nonzero entries, $\beta_j^* = s_j j^{-1/2}$ for $j = 1, \ldots, 10$, with fixed random signs $s_j \in \{-1, 1\}$. Mismatch directions are j^{-1}, constant, and $(j/p)^2$ for the low-frequency, white, and high-frequency arms, respectively. Each direction is normalized to $\|\delta\| = 0.5$ and negated when necessary to make $\beta^{*\top}\delta \leq 0$. Target and auxiliary labels are $x^\top\beta^* + \varepsilon_r$ and $x^\top(\beta^* + \delta) + \varepsilon_h$, with noise variances 1 and 4. We use $n_h = 6n_r$, $n_r \in \{50, 100, 200, 500, 1000, 2000\}$, 200 replicates for the low-frequency arm, and 50 for each other arm.

Validation and test separation. Each target replicate is split in random order into 80% fitting and 20% validation samples. Candidate values are $\mu \in \{0.001, 0.01, 0.05, 0.1, 0.2\}$ and $w \in \{0, 0.05, 0.1, 0.2, 0.4, 0.5, 0.7, 1\}$. For every w, μ is selected by minimizing mean squared error between model predictions and the noisy target validation labels. The selected w minimizes the same validation-label criterion across the resulting optimally tuned candidates; equivalently, the reported selection is the joint minimum over the (μ, w) grid. Target-only and fixed-$w = 0.5$ baselines receive their own validation-selected μ. Neither β^* nor test data enters any selection step. After selection, β^* is used only to compute noise-free target excess risk on the held-out CIFAR test features. Separately labelled oracle test-grid gains compare independently test-minimized target-only and mixed grids; they are diagnostic upper bounds, not validation-selected decisions.

CIFAR-MLP calibration. The additional low-frequency MLP arm uses $n_r \in \{100, 500, 1000\}$, width 32, 500 Adam steps, and $L_2 \in \{0, 10^{-4}, 10^{-3}\}$. Target-only L2 calibration averages noisy validation-label MSE over two seeds, using the same 80/20 split convention, and final test risks are reported over five evaluation seeds. Test features and β^* do not participate in L2 selection.

Aggregation conventions. For the coarse synthetic sweeps above, benefits are computed on the estimator's risk curve $(R(0) - \min_\lambda R(\lambda))$, not by taking per-replicate minima before averaging (which incurs a Jensen bias); single-seed artifacts are flagged per cell in the JSON outputs. For CIFAR, validation selection is instead performed independently within each replicate before averaging gains, matching the deployable decision procedure; error bars show 1.96 standard errors.

Compute. CPU only; closed-form 200-replicate sweeps take minutes; the full coarse batch of Adam-trained arms takes a

few hours on one workstation.

References